Bioinformatics



KB8029

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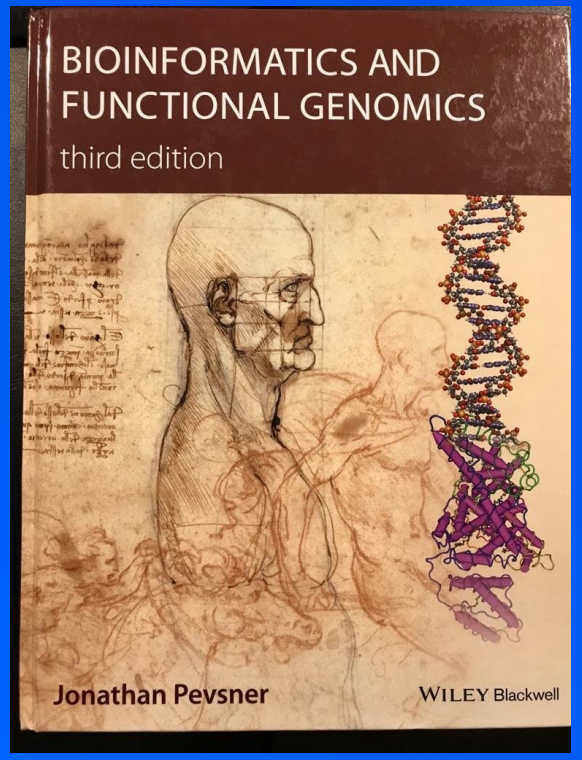
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Stockholm University and SciLife Lab, February 27, 2026

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Protein Structure



Contents in Brief

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PART II Genomewide Analysis of DNA, RNA, and Protein

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Protein Structure

CHAPTER 13

A visitor to the Accademia in Florence can see magnificent images that emerged from blocks of marble at the hands of Michelangelo. By analogy, the noncrystallographer can capture the vision that a crystallographer has when admiring a rigorously shaped crystal before exploring the marvelous structure hidden within. So the Protein Data Bank is our museum, with models of molecules reflecting the wonders of nature and complex shapes that may be as old as life itself. With the aid of interactive graphics and networking, the PDB makes these images readily available. What wonders still remain hidden as we build, compare, and extend our database?

Edgar F. Meyer (1997)

Plus, some additional material

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Highlights:

From atoms to proteins

Backbone and side chains

Amino acids and their properties (primary structure)

Secondary structure elements

Tertiary and quaternary structure

Comparing

tertiary structure

quaternary structure

Predicting

tertiary structure

quaternary structure

Classifying and storing

Protein domains

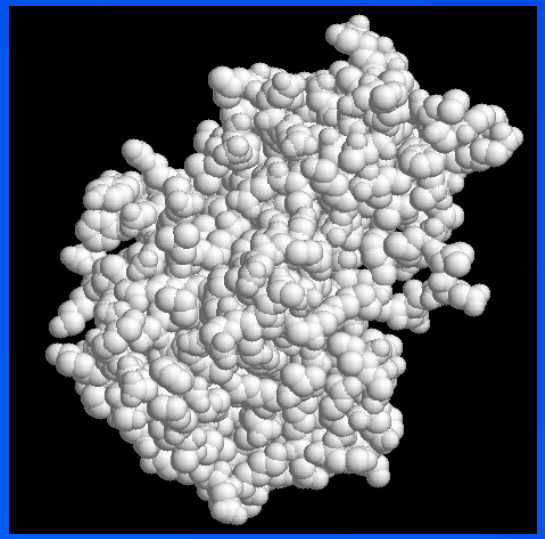
Main databases

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Proteins

are large molecules

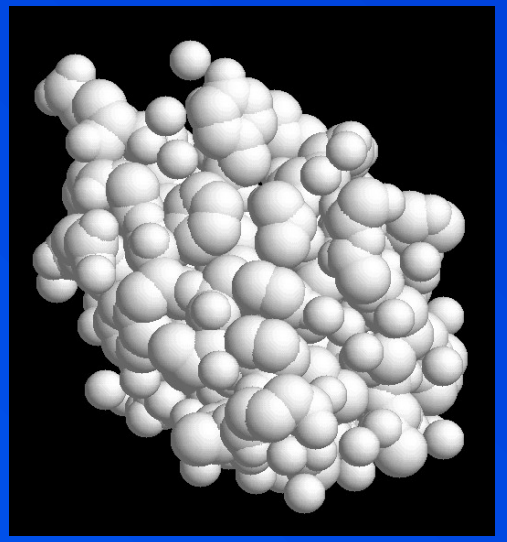
Alcohol dehydrogenase



65 Å

2643 atoms

Insulin



34 Å

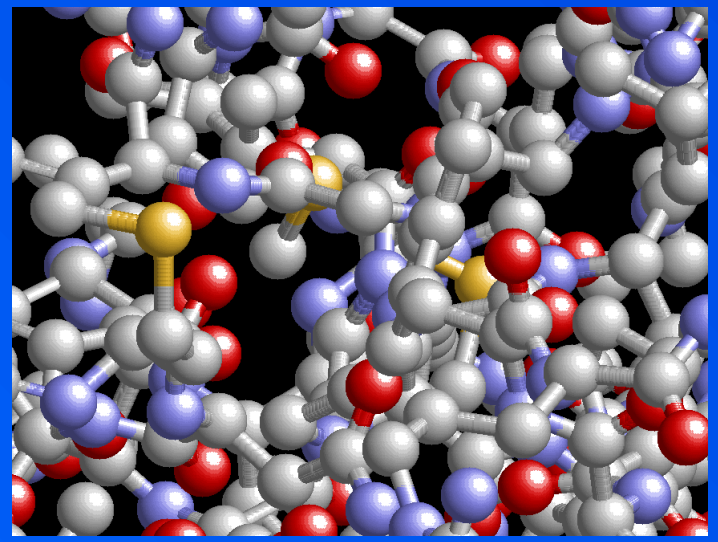
411 atoms

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Proteins

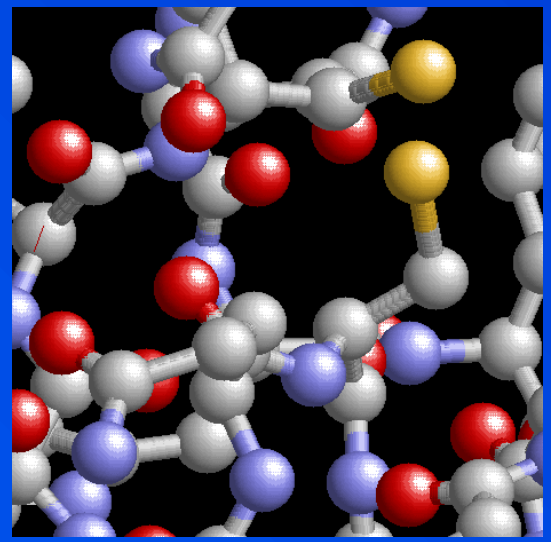
are large molecules

Alcohol dehydrogenase



2643 atoms

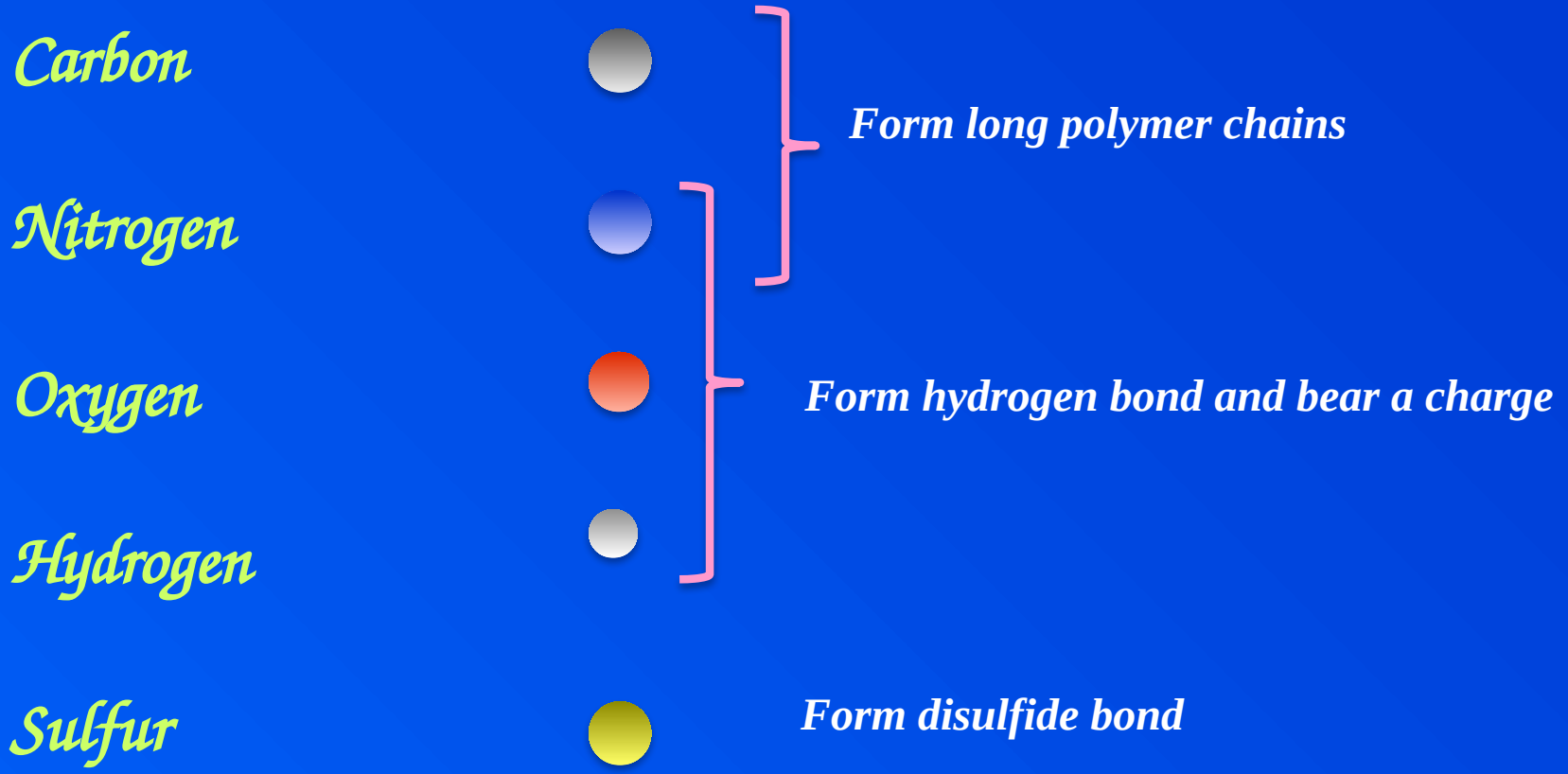
Insulin



411 atoms

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Atoms in Proteins



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Backbone

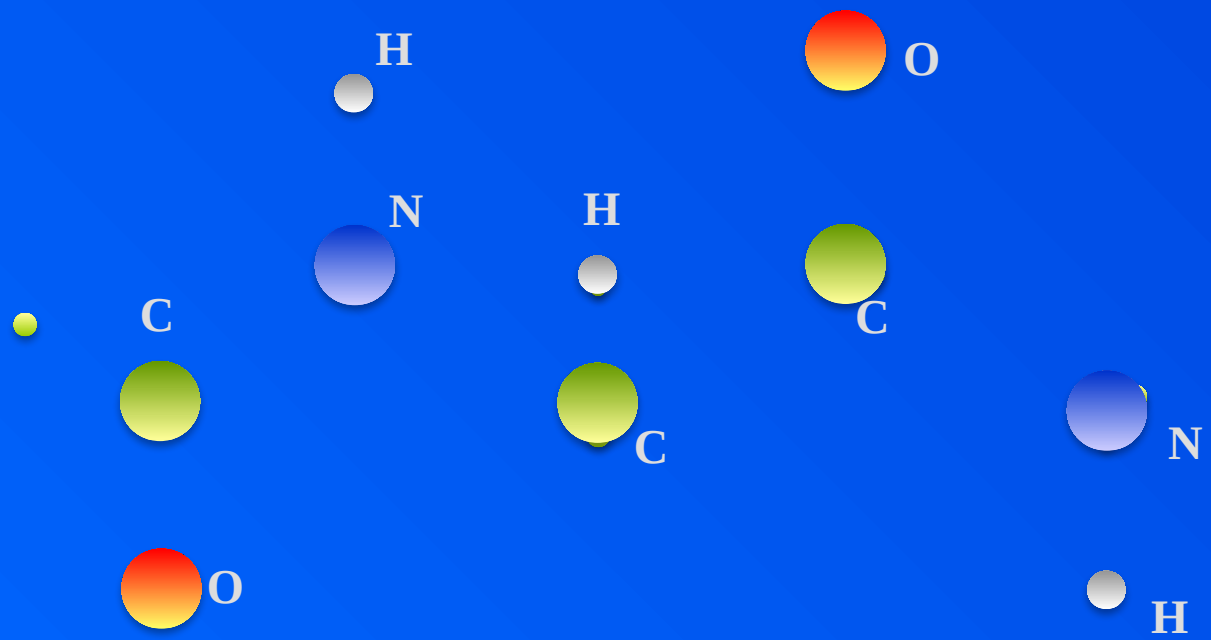
Proteins are flexible polymer chains

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Backbone

Proteins are flexible polymer chains polymer

Generic structure

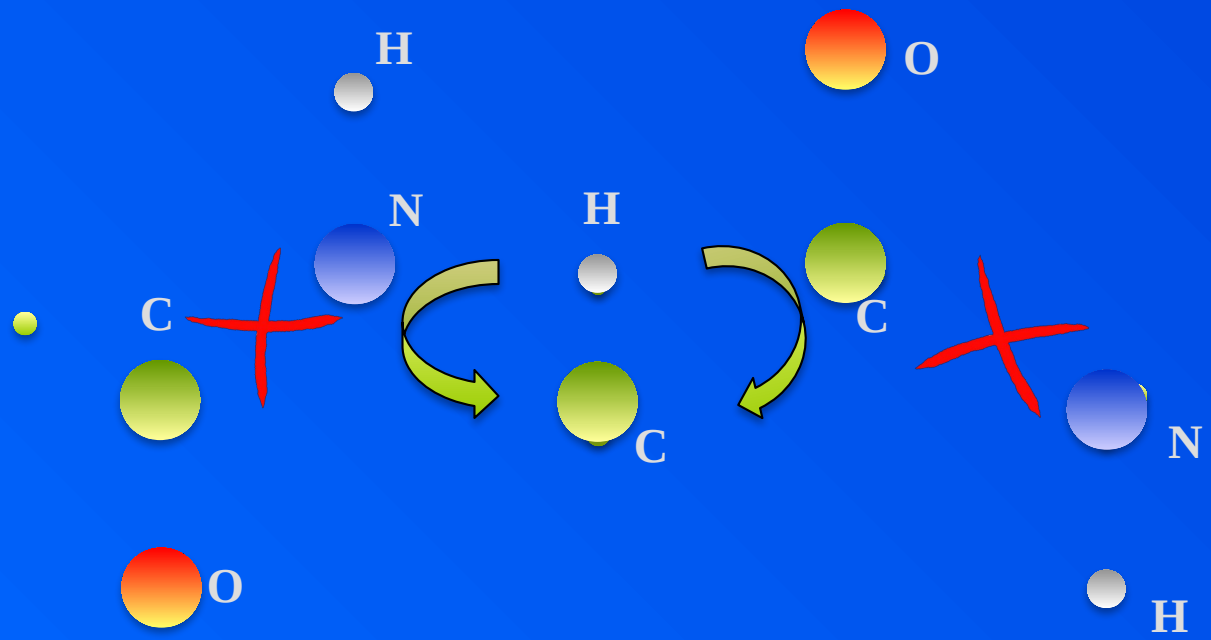


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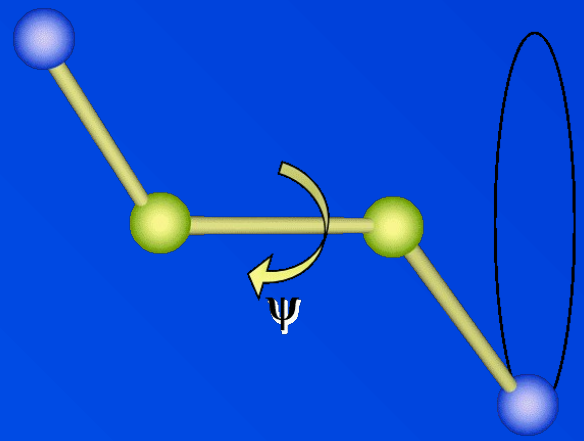
Backbone

Proteins are flexible polymers of amino acids

Generic structure



Dihedral angle

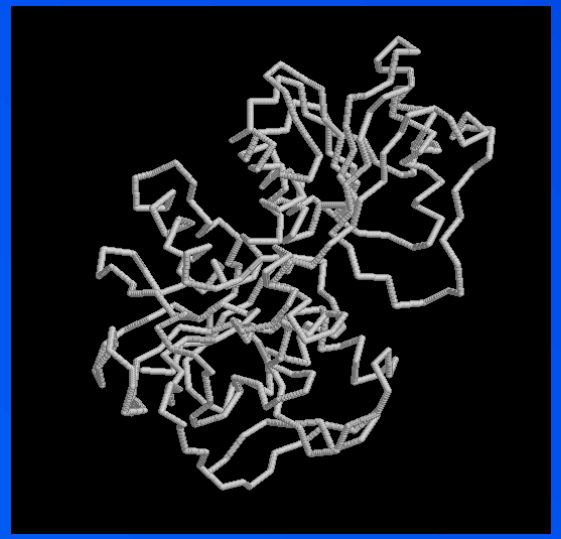


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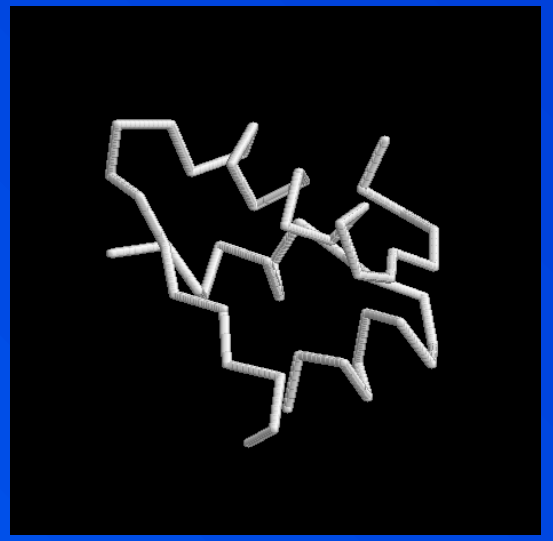
Proteins

Backbone

Alcohol dehydrogenase



Insulin





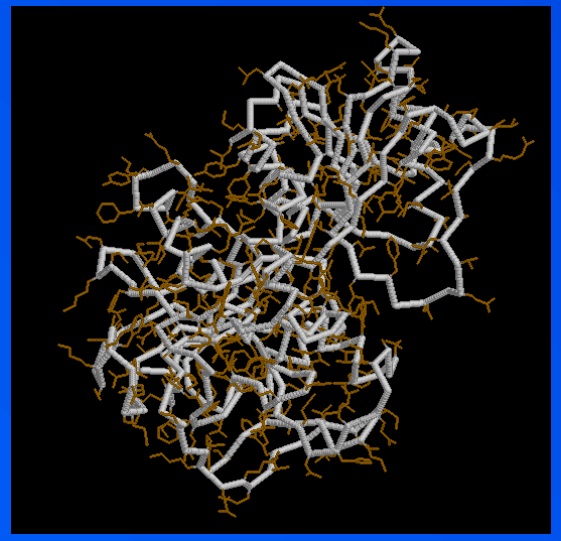
Backbone is just a skeleton of a protein

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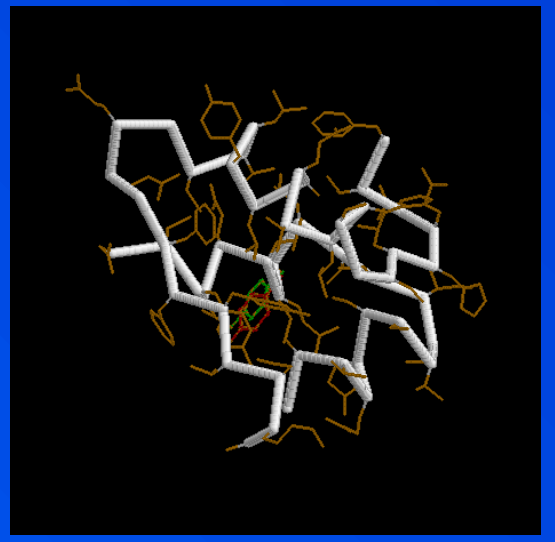
Proteins

Backbone and side chains

Alcohol dehydrogenase



Insulin





Side chains are “flesh on the bones”

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Side chains

Hydrophobic amino acids



Alanine (ALA, A) *Leucine* (LEU, L) *Isoleucine* (ILE, I) *Methionine* (MET, M) *Phenylalanine* (PHE, F) *Proline* (PRO, P) *Tryptophan* (TRP, W) *Valine* (VAL, V) *Glycine* (GLY, G)

Polar amino acids



Asparagine (ASN, N) *Glutamine* (GLN, Q) *Serine* (SER, S) *Threonine* (THR, T) *Tyrosine* (TYR, Y) *Histidine* (HIS, H) *Cysteine* (CYS, C)

Charged (titratable) amino acids

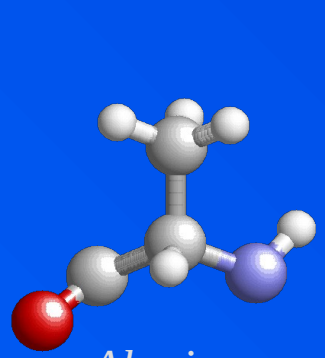


Aspartic acid (ASP, D) *Glutamic acid* (GLU, E) *Lysine* (LYS, K) *Arginine* (ARG, R)

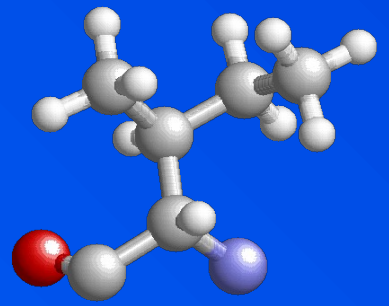
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Hydrophobic side chains

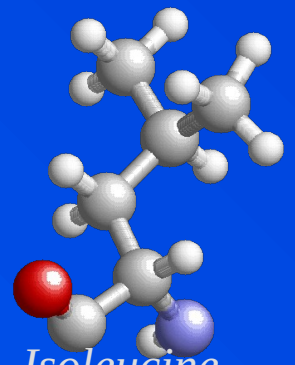
Incapable of forming hydrogen bond



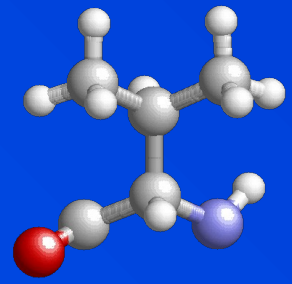
Alanine
(ALA, A)



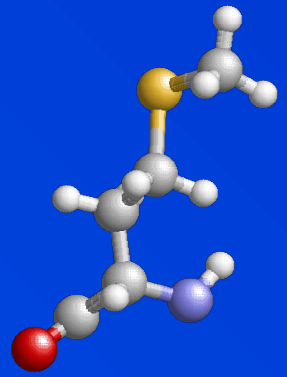
Leucine
(LEU, L)



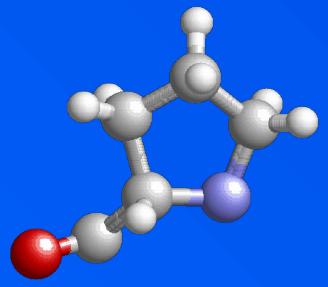
Isoleucine
(ILE,I)



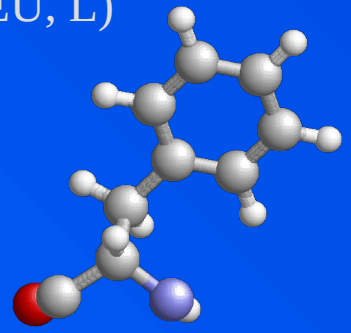
Valine
(VAL, V)



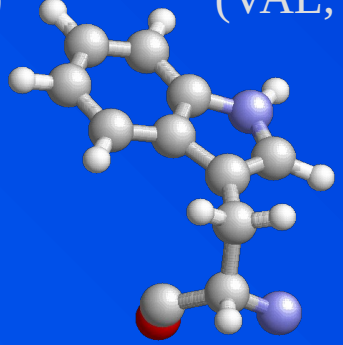
Methionine
(MET, M)



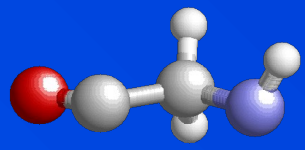
Proline
(PRO, P)



Phenylalanine
(PHE, F)



Tryptophan
(TRP, W)

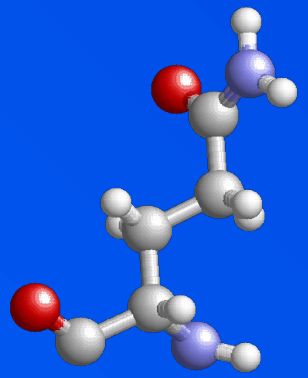


Glycine
(GLY, G)

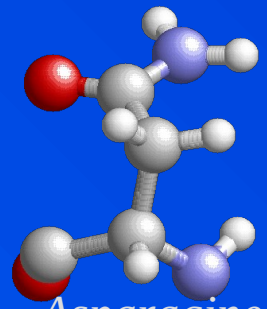
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Polar side chains

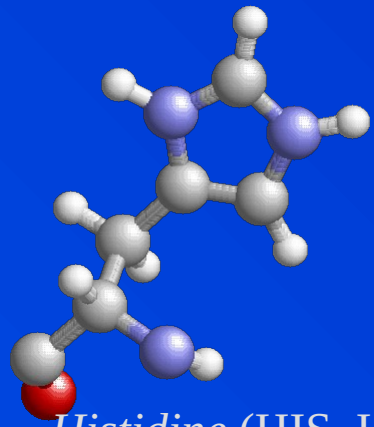
Capable of forming hydrogen bond



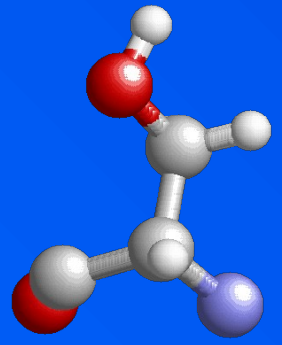
Glutamine (GLU, Q)



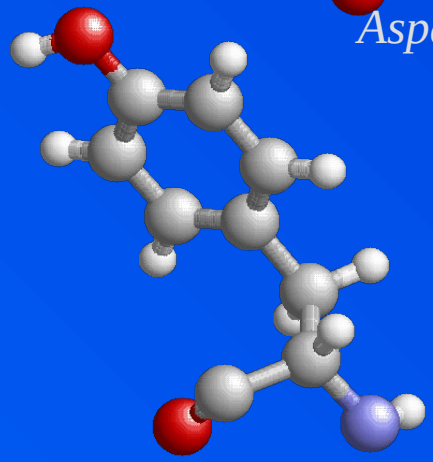
Asparagine (ASN, N)



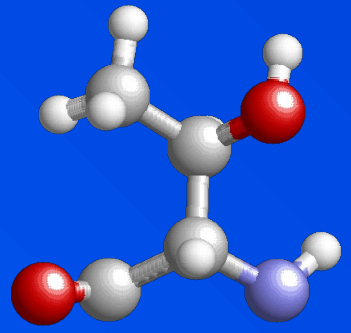
Histidine (HIS, H)



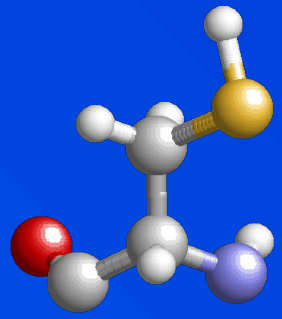
Serine (SER, S)



Tyrosine (TYR, Y)



Threonine (THR, T)



Cysteine (CYS, C)

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Charged side chains

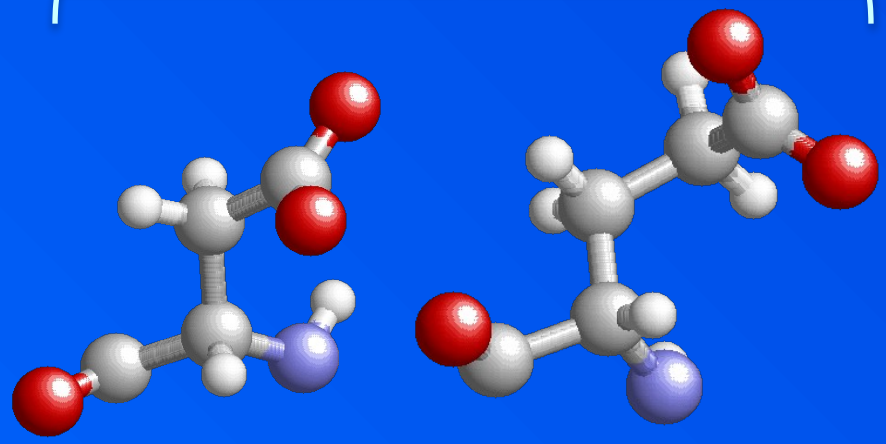
May bear charge

Acidic residues (negative charge)

Basic residues (positive charge)

Not protonated form

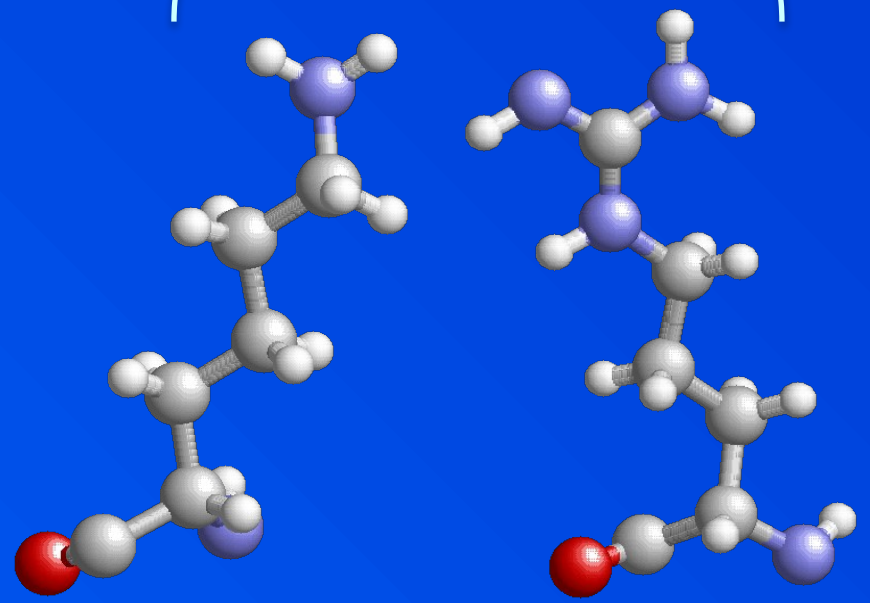
charged



Aspartic acid (ASP, D)

Glutamic acid (GLU, E)

neutral



Lysine (LYS, K)

Arginine (ARG, R)

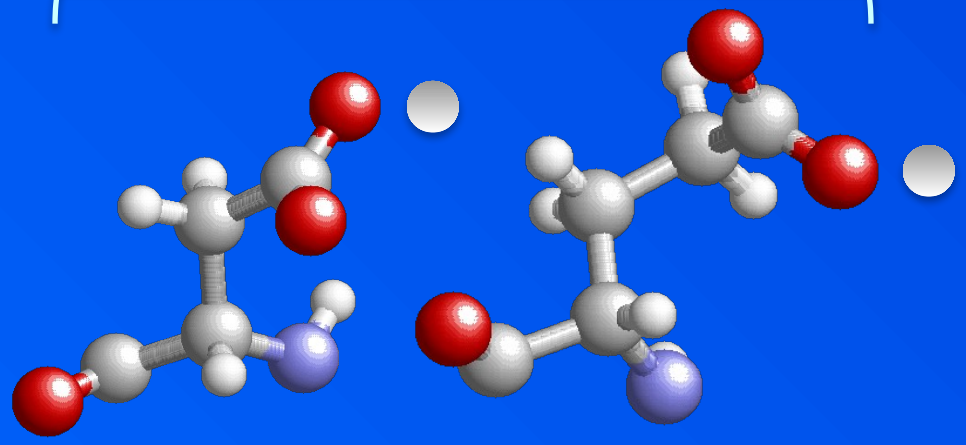
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Charged side chains

May bear charge

Acidic residues (negative charge)

neutral



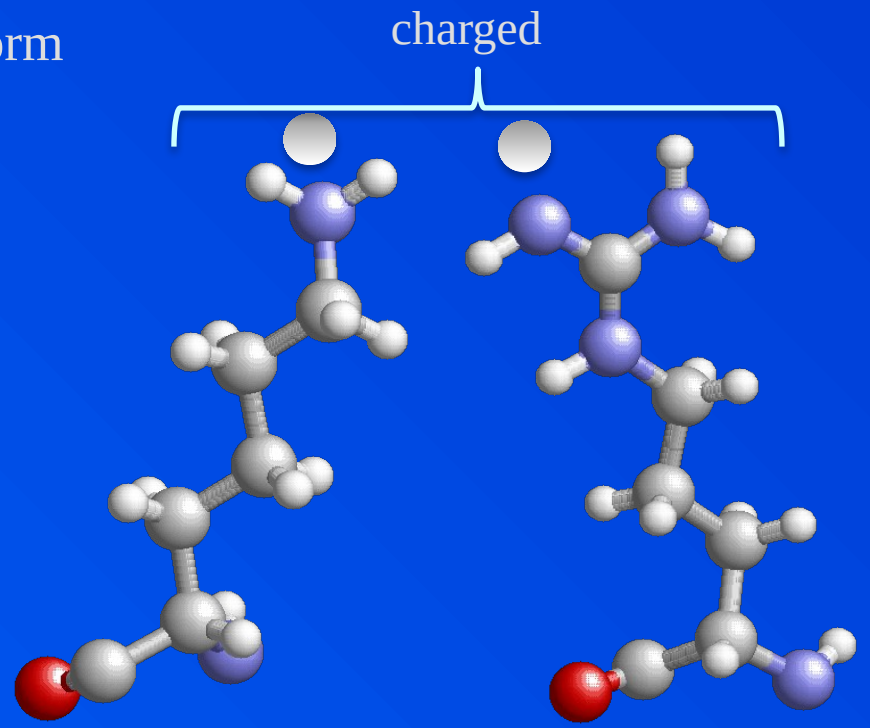
Aspartic acid (ASP, D)

Protonated form

Glutamic acid (GLU, E)

Basic residues (positive charge)

charged



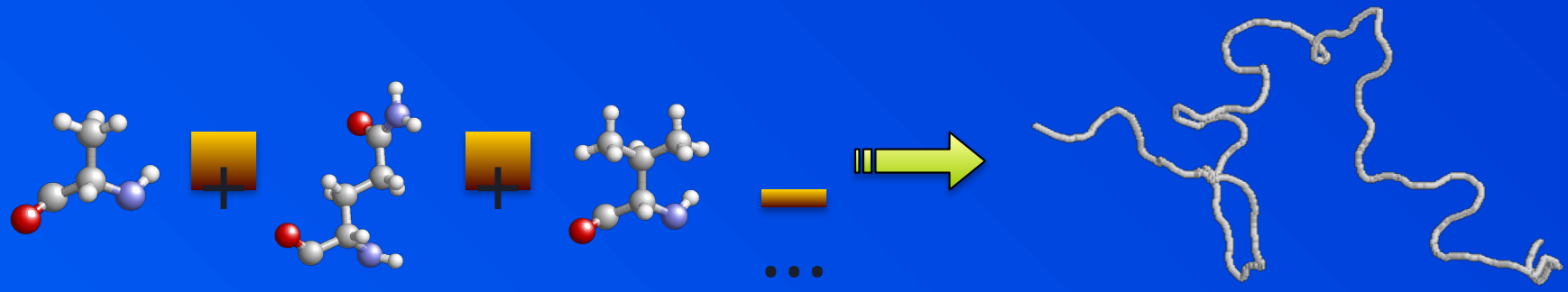
Lysine (LYS, K)

Arginine (ARG, R)

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Primary structure

Order of amino acids in polypeptide chain



Amino acids

Flexible polypeptide chain (not a protein yet)

Polypeptide chain in water becomes a protein



Polypeptide chain

Water

Protein

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Secondary structure

Rigid elements of protein structure

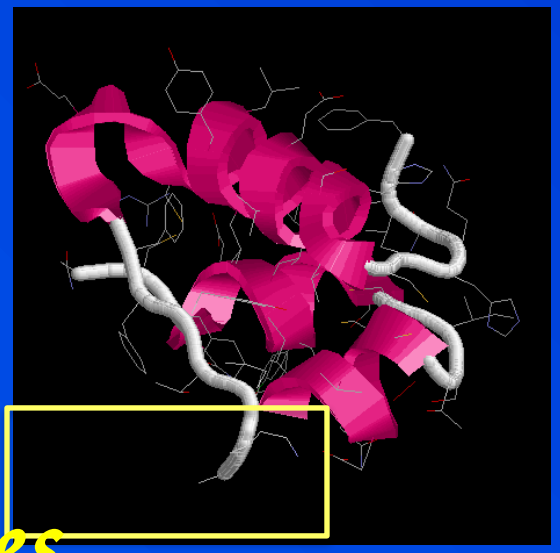
Alcohol dehydrogenase



beta strands

alpha helices

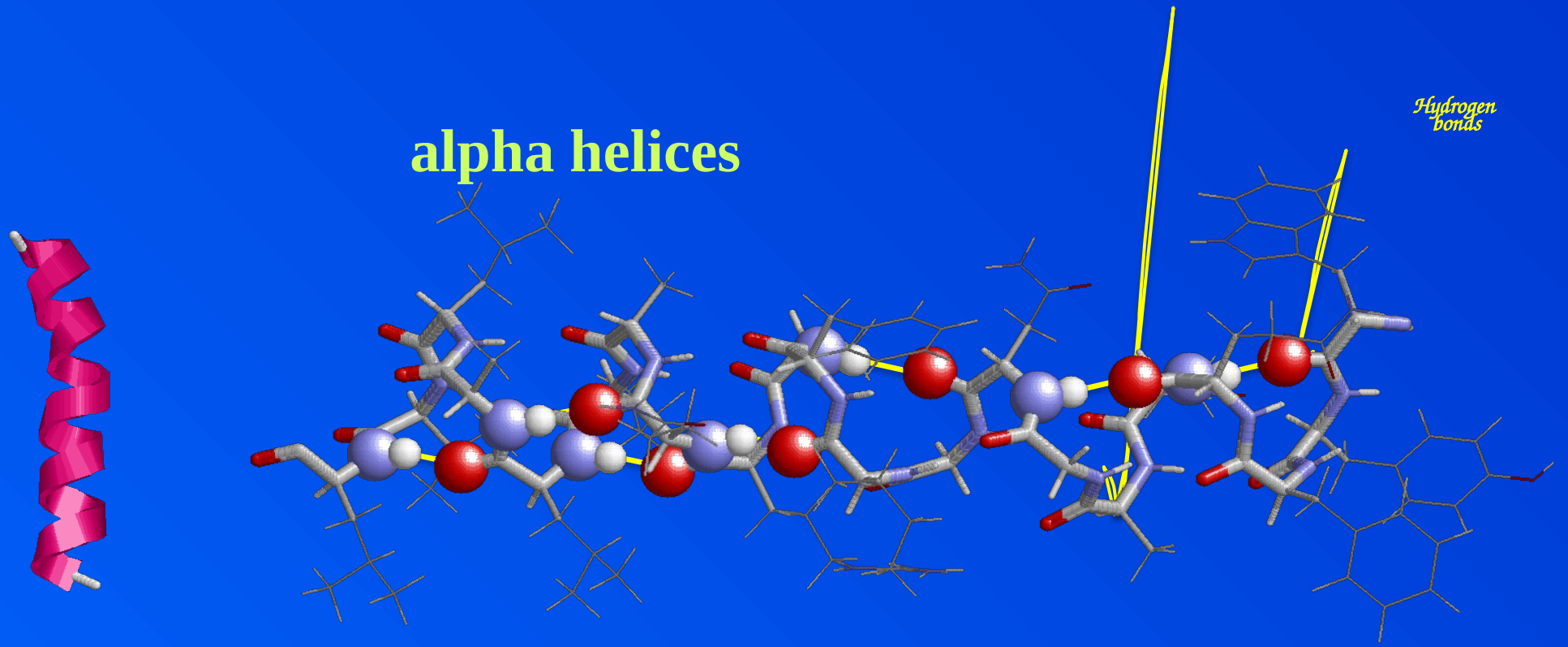
Insulin



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Secondary structure

alpha helices



Stabilized by hydrogen bonds between NH and O atoms of the backbone separated by 4 residues

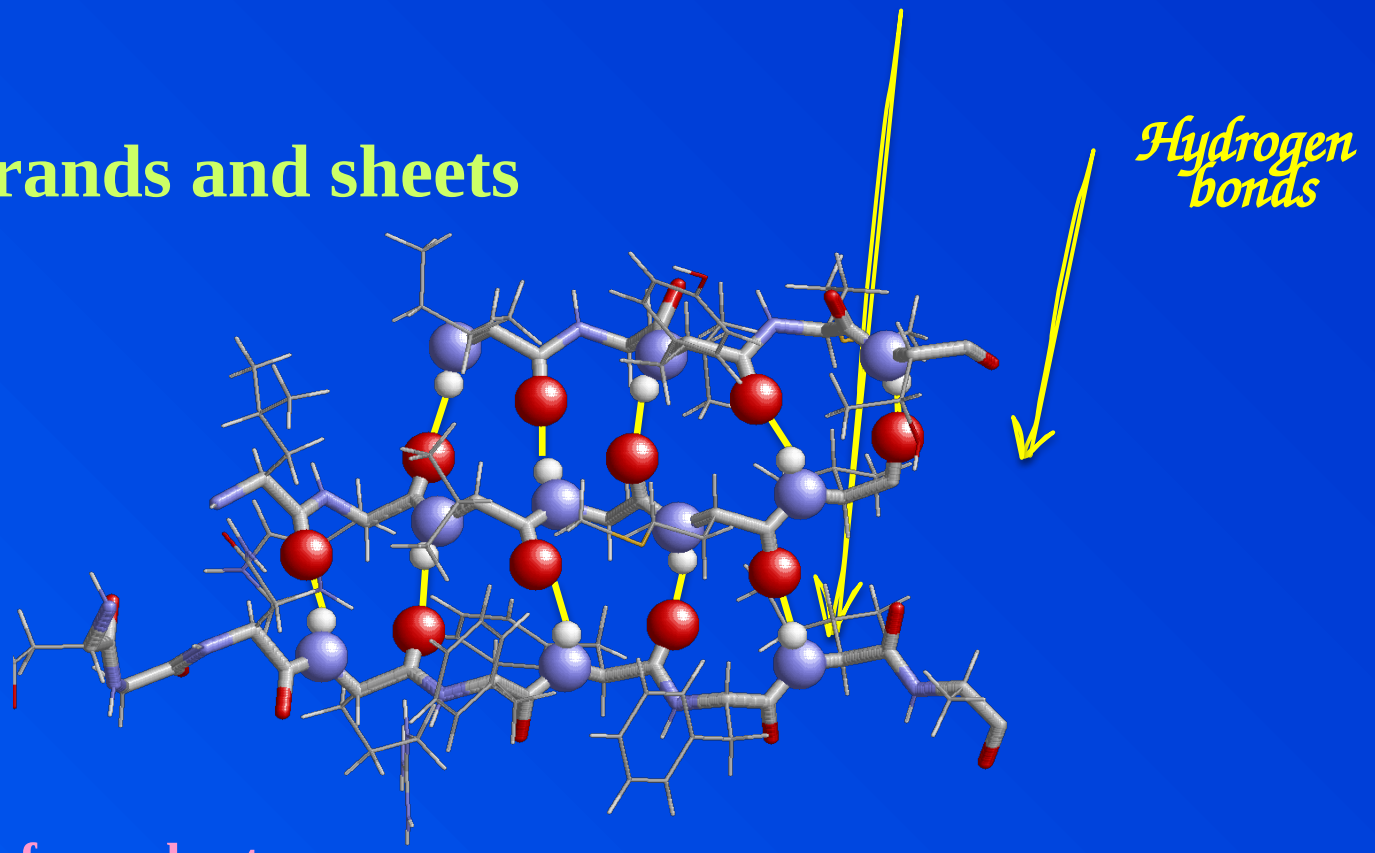
Side chains do not participate in stabilizing helix

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Secondary structure

beta strands and sheets





Strands do not exist alone and form sheets

Stabilized by hydrogen bonds between N and O atoms in the neighboring strands

Side chains do not participate in stabilizing sheets

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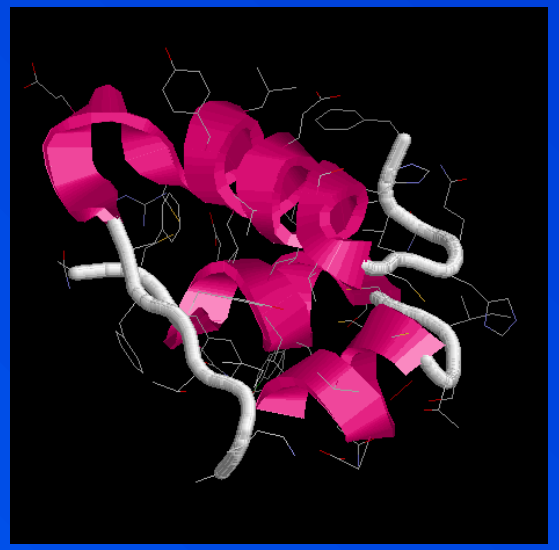
Tertiary structure

Mutual arrangement of secondary structure elements

Alcohol dehydrogenase



Insulin



Stabilized by hydrophobic effect

Stabilized by non-covalent bonds between atoms of both backbone and side chains

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Non-covalent interactions

van der Waals interactions

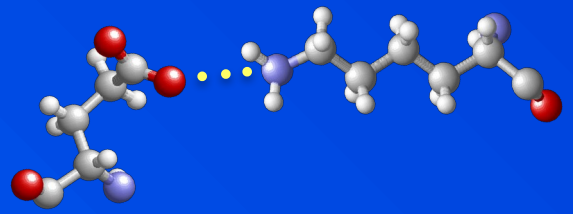
Attractive long-range part

Repulsive short-range part



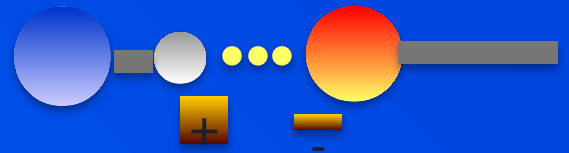
Electrostatic interactions

Salt bridge

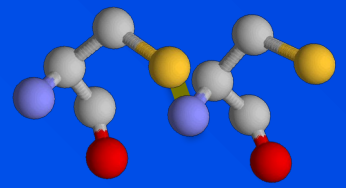


Hydrogen bonding

Dipole-dipole interactions



Disulfide bond



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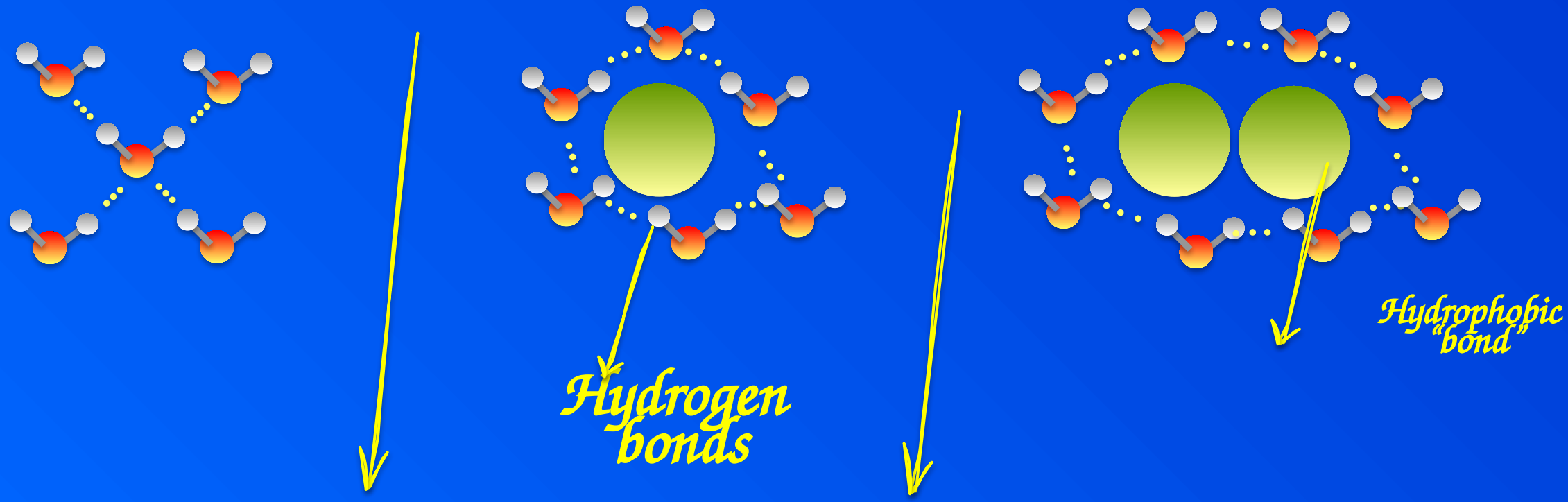
Hydrophobic effect

Proteins function in water

Pure water

One hydrophobic atom

Two hydrophobic atoms



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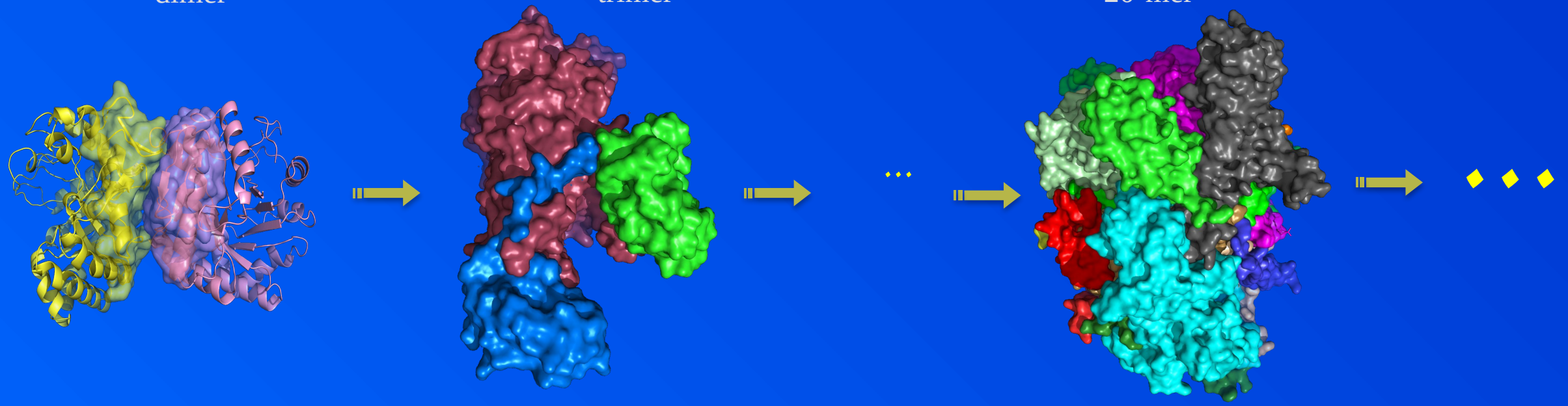
Quaternary structure

Mutual arrangement of protein molecules

dimer

trimer

20-mer



Stabilized by hydrophobic effect

Stabilized by non-covalent bonds between atoms from different chains

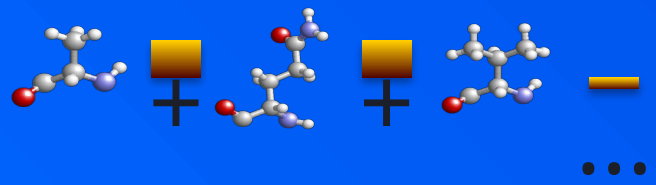
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Protein structure

Summary

Primary

MGLSDGEWQLVLNVWGKVEADIPGHGQEVLIR
LFKGHPETLEKFDKFKHLKSEDEMKASEDLKK
HGATVLTALGGILKKKGHHEAEIKPLAQSHAT
KHKIPVKYLEFISECIIQVLQSKHPGDFGADA
QGAMNKALELFRKDMASNYKELGFQG



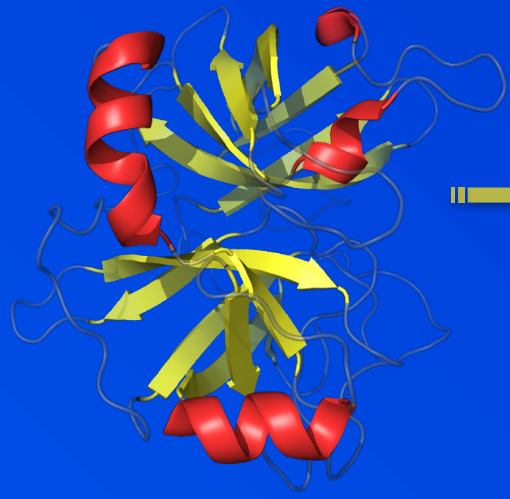
amino acid sequence

Secondary



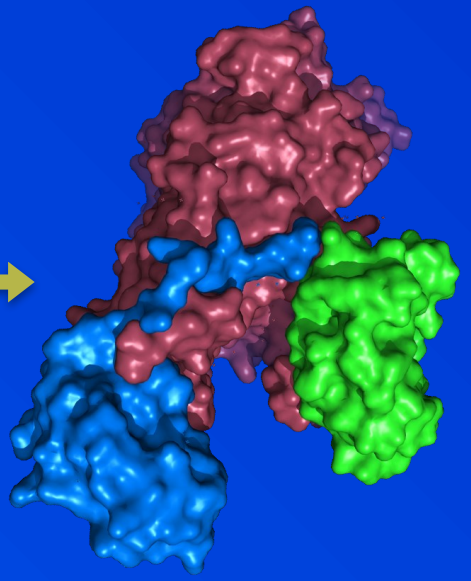
alpha-helices & beta strands/sheets

Tertiary



spatial arrangement of secondary structure elements (single protein)

Quaternary



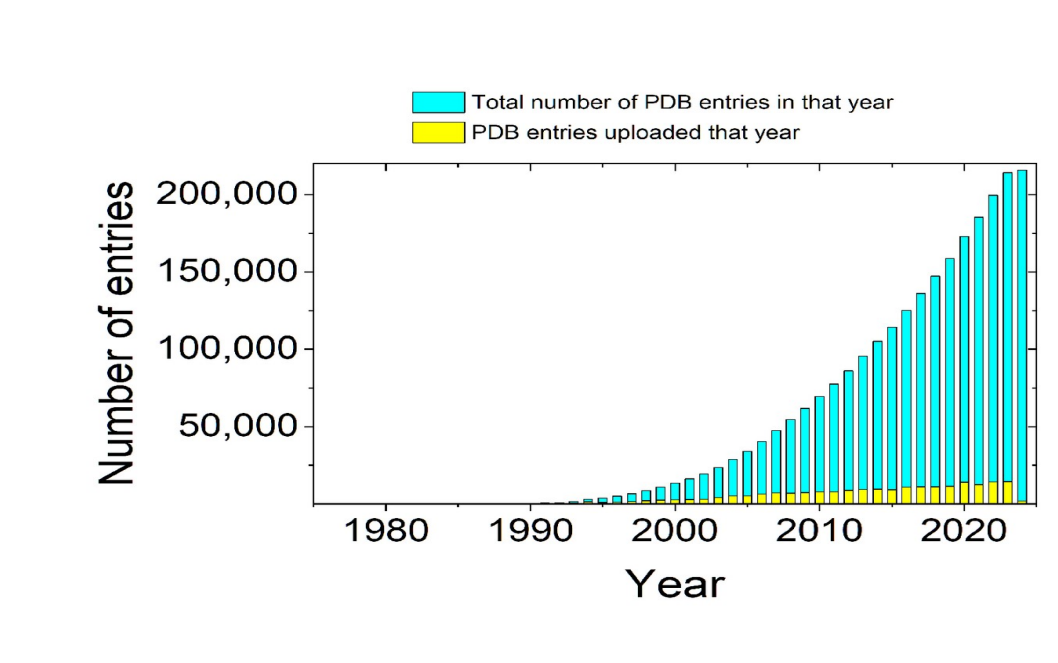
spatial arrangement of single proteins

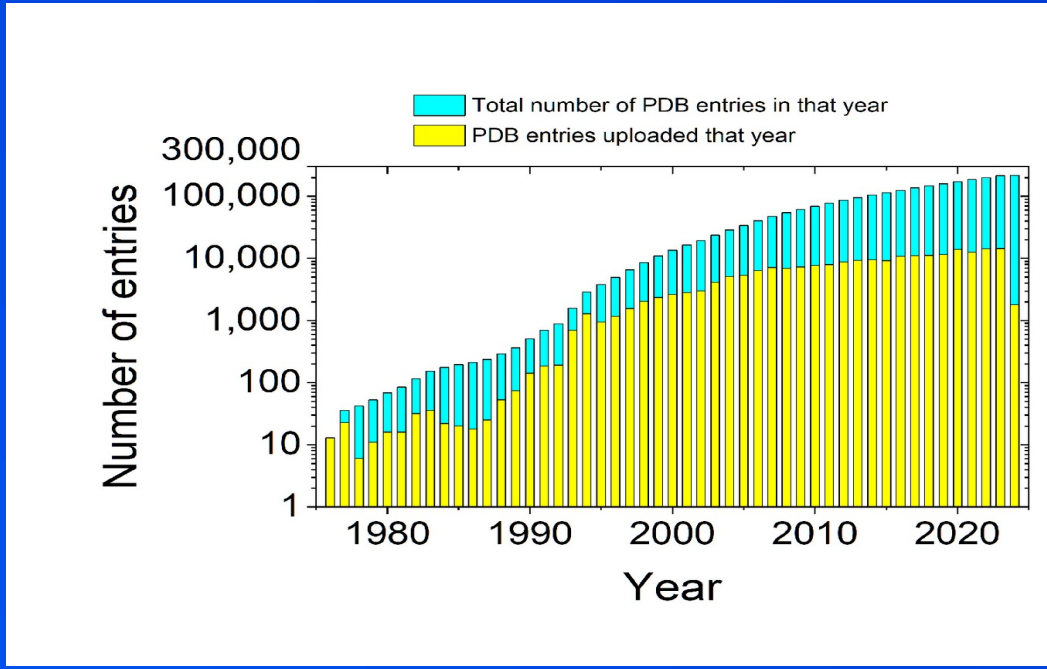
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Structural databases

Protein Data Bank (PDB)

Number of entries





Alpha Fold model database

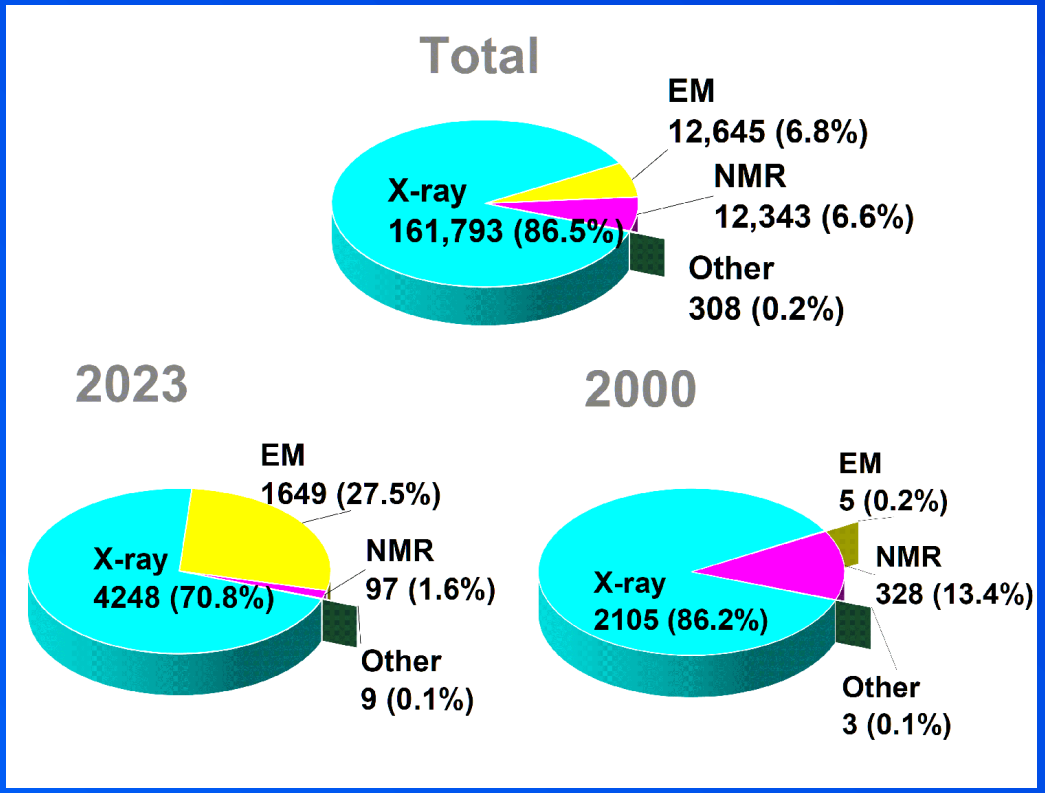
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Structural databases

Protein Data Bank (PDB)

RCSB PDB PROTEIN DATA BANK

Breakdown of PDB entries by experimental techniques



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Comparing Tertiary Structures

Rigid body transformation

Various measures of structure similarity

Root Mean Square Distance

TM-score

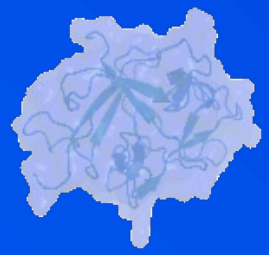
Global Distance Test, GDT score

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Rigid-body transformation

Old coordinates

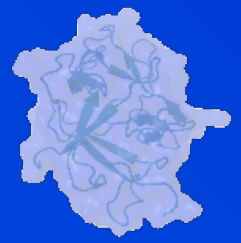
$\overline{Z_{old}} = (x_{old}, y_{old}, z_{old})$





New coordinates

$\overline{Z_{new}} = (x_{new}, y_{new}, z_{new})$



Rigid-body transformation

Old coordinates

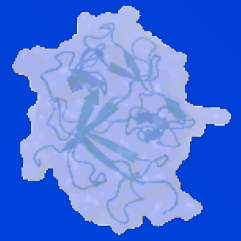
$\overline{Z_{old}} = (x_{old}, y_{old}, z_{old})$

New coordinates

$\overline{Z_{new}} = (x_{new}, y_{new}, z_{new})$

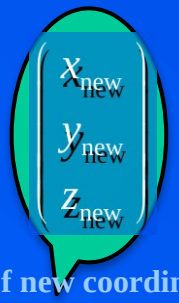




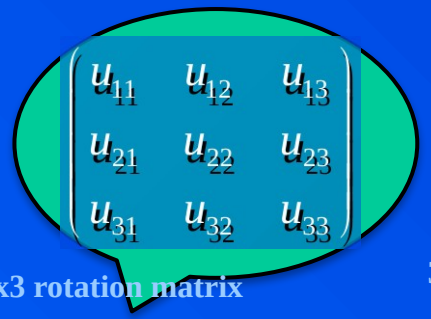


Matrix-vector form:

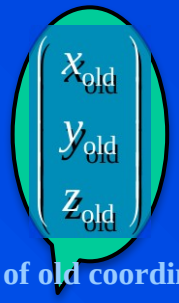
$$\vec{c}_{new} = U \times \vec{c}_{old} + \vec{T}$$



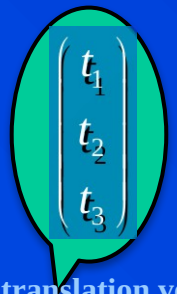
3D vector of new coordinates



3x3 rotation matrix



3D vector of old coordinates



3D translation vector

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Rigid-body transformation

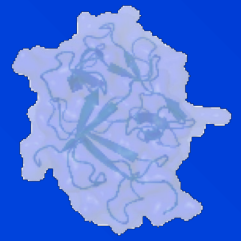
Old coordinates

$\overline{Z_{old}} = (x_{old}, y_{old}, z_{old})$

New coordinates

$\overline{Z_{new}} = (x_{new}, y_{new}, z_{new})$





Matrix-vector form:

$$\vec{c}_{new} = U \times \vec{c}_{old} + \vec{T}$$

$\begin{pmatrix} x_{new} \\ y_{new} \\ z_{new} \end{pmatrix}$ 3D vector of new coordinates

Ouput of programs

$\begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix}$ 3x3 rotation matrix

$\begin{pmatrix} x_{old} \\ y_{old} \\ z_{old} \end{pmatrix}$ 3D vector of old coordinates

Ouput of programs

$\begin{pmatrix} t_1 \\ t_2 \\ t_3 \end{pmatrix}$ 3D translation vector

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Rigid-body transformation

Old coordinates

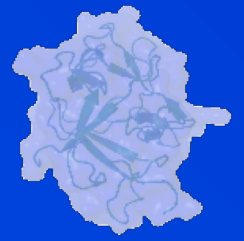
$\overline{Z_{old}} = (x_{old}, y_{old}, z_{old})$

New coordinates

$\overline{Z_{new}} = (x_{new}, y_{new}, z_{new})$



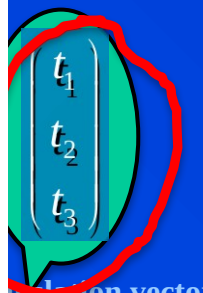




Matrix-vector f

ms

3D vector of ne



nslation vector

```
apollo:~/DOCKGROUND/UNBOUND4.0/PDB> TMalign_32 1akj_b1.pdb 1akj_u1.pdb

 **************************************************************************
 *                                TM-align                                *
 * A protein structural alignment algorithm based on TM-score             *
 * Reference: Y. Zhang and J. Skolnick, Nucl. Acids Res. 2005 33, 2302-9  *
 * Comments on the program, please email to: yzhang@ku.edu                *
 **************************************************************************

Chain 1:1akj_b1.pd  Size= 276
Chain 2:1akj_u1.pd  Size= 100 (TM-score is normalized by  100)

Aligned length=  96, RMSD=  2.44, TM-score=0.81293, ID=0.198

 ------- rotation matrix to rotate Chain-1 to Chain-2 ------
 i          t(i)         u(i,1)         u(i,2)         u(i,3)
 1     47.2222036078  -0.6678570614   0.5034442963  -0.5481886410
 2     93.8991687527  -0.5918602971  -0.8058170138  -0.0189823354
 3     -7.7865135482  -0.4512962821   0.3117736051   0.8361392736

(":" denotes the residue pairs of distance < 5.0 Angstrom)
GSHSMRYFFTSVSRPGRGEPRFIAVGYVDDTQFVRFDSDAASQRMEPRAPWIEQEGPEYWDGETRKVKAHSQTHRVDLGTLRGYYNQSEAGSHTVQRMYG
GYHQYAYDGKDYIALKEDLRSWTAADMAAQTTKHKWEAAHVAEQLRAYLEGTCVEWLRRYLENGKETLQRTDAPKTHMTHHA-VS-DHEATLRCWALSFY
GEDQTQDTELVETRPAGDGTFQKWAAVVVPSGQEQRYTCHVQHEGLPKPLTLRWEP--

                                                                                                    :::::::::::::: .. ::::::::::::::
::::: ::::::::::::::::::::::::::..:::::::::::::::::::::.
------------------------------------------------------------------------------------------------------------------------
------------------------------------------------------------------MIQRTPKIQVYSRHPAENGKSNFLNCYVSGFH
GERIE-KVEHSDLSFSKDWSFYLLYYTEFTPTEKDEYACRVNHVTLSQPKIVKWDRDM
```

Rotation matrix

Translation vector

Rigid-body transformation

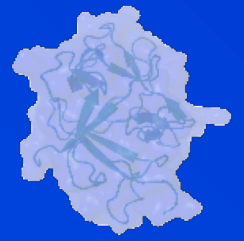
Old coordinates

$\overline{Z_{old}} = (x_{old}, y_{old}, z_{old})$

New coordinates

$\overline{Z_{new}} = (x_{new}, y_{new}, z_{new})$





Matrix-vector form:

$$\vec{c}_{new} = \mathbf{U} \times \vec{c}_{old} + \vec{T}$$

$$\begin{pmatrix} x_{new} \\ y_{new} \\ z_{new} \end{pmatrix}$$

3D vector of new coordinates

Ouput of programs

$$\begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix}$$

3x3 rotation matrix

$$\begin{pmatrix} x_{old} \\ y_{old} \\ z_{old} \end{pmatrix}$$

3D vector of old coordinates

Ouput of programs

$$\begin{pmatrix} t_1 \\ t_2 \\ t_3 \end{pmatrix}$$

3D translation vector

Algebraic form:

$$x_{new} = u_{11} x_{old} + u_{12} y_{old} + u_{13} z_{old} + t_1$$

$$y_{new} = u_{21} x_{old} + u_{22} y_{old} + u_{23} z_{old} + t_2$$

$$z_{new} = u_{31} x_{old} + u_{32} y_{old} + u_{33} z_{old} + t_3$$

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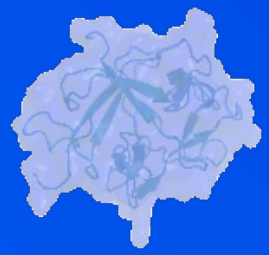
Rigid-body transformation

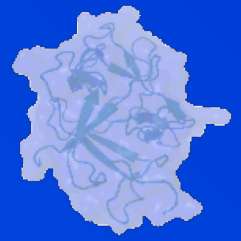
Old coordinates

$\overline{Z_{old}} = (x_{old}, y_{old}, z_{old})$

New coordinates

$\overline{Z_{new}} = (x_{new}, y_{new}, z_{new})$

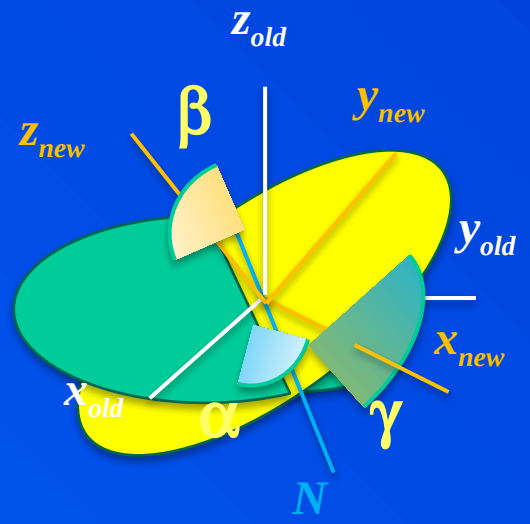


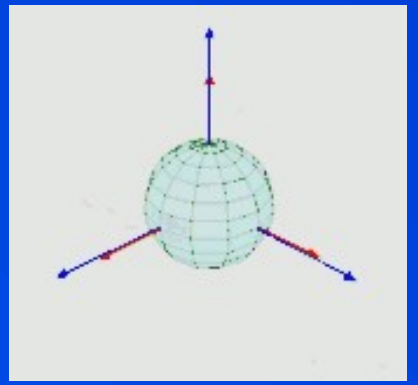


Classical Euler angles (α, β, γ):

Rotation matrix in terms of classical Euler angles

(first by α, second by β, third by γ, in a right-hand system):





$$U = \begin{pmatrix} \cos\gamma\cos\alpha - \cos\beta\sin\alpha\sin\gamma & \cos\gamma\sin\alpha + \cos\beta\cos\alpha\sin\gamma & \sin\gamma\sin\beta \\ -\sin\gamma\cos\alpha - \cos\beta\sin\alpha\cos\gamma & -\sin\gamma\sin\alpha + \cos\beta\cos\alpha\cos\beta & \cos\gamma\sin\alpha \\ \sin\beta\sin\alpha & -\sin\beta\cos\alpha & \cos\beta \end{pmatrix}$$

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Comparing Tertiary Structures

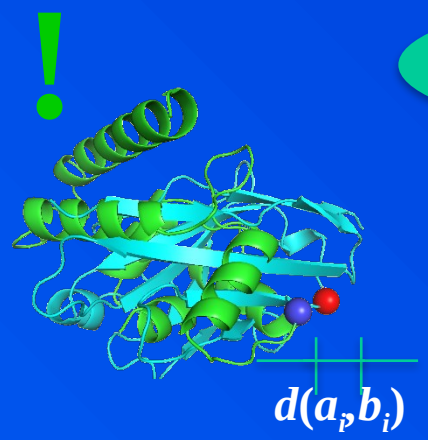
Root-mean-square distance (RMSD)

$$\mathrm{RMSD}(A,B) = \sqrt{\frac{1}{\min(N,M)} \sum_{i=1}^{\min(N,M)} d^2(a_i, b_i)}$$

Distance between *corresponding* points



Not in original A and B positions!

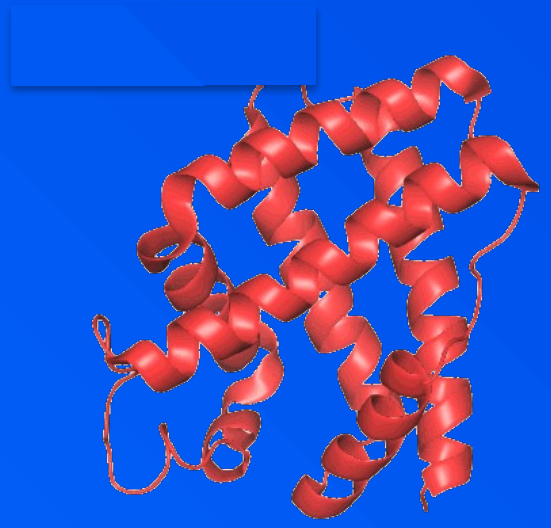


Align A and B to minimize RMSD!

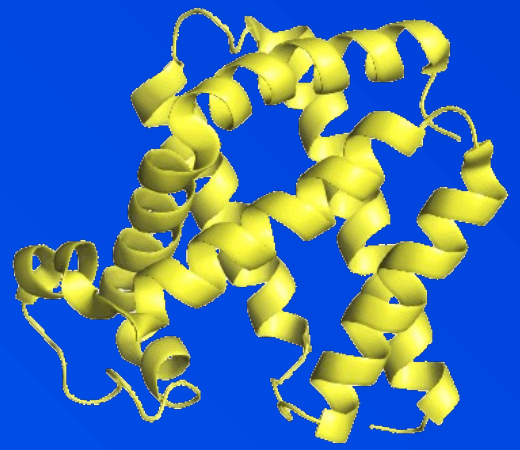
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Alignment of Structures

Re-arranging structures in space to infer regions of similarity



1LH1, leghemoglobin from yellow lupin



1URV, cytoglobin from human

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Comparing Tertiary Structures

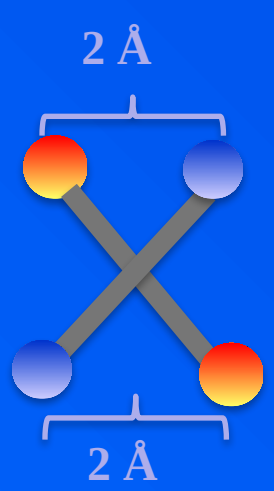
Why

$$RMSD(A,B) = \sqrt{\frac{1}{\min(N,M)} \sum_{i=1}^{\min(N,M)} d^2(a_i,b_i)}$$

and not average distance

$$\bar{d}(A,B) = \frac{1}{\min(N,M)} \sum_{i=1}^{\min(N,M)} d(a_i,b_i)$$

?





RMSD represents sample standard deviation

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Comparing Tertiary Structures

TM-score (TM-align, MM-align, I-align)

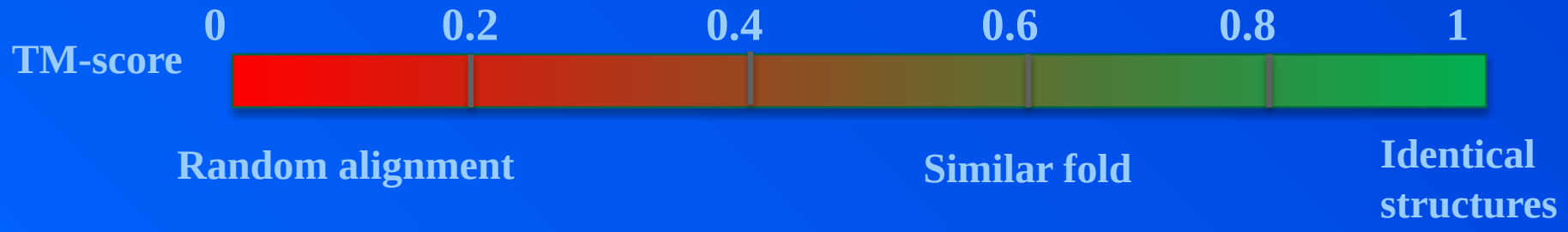
Length of the optimal alignment

Length of the target protein

Distance between *i*-th pair of aligned residues

$$\text{TM-score} = \frac{1}{L_{tar}} \sum_{i=1}^{L_{ali}} \frac{1}{1 + \left(\frac{d_i}{d_0(L_{tar})} \right)^2}$$

$$d_0(L_{tar}) = 1.24 \sqrt[3]{L_{tar} - 15} - 1.8$$



Zhang & Skolnick (2005) Nucleic Acids Res., 33:2302-2309

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Comparing Tertiary Structures

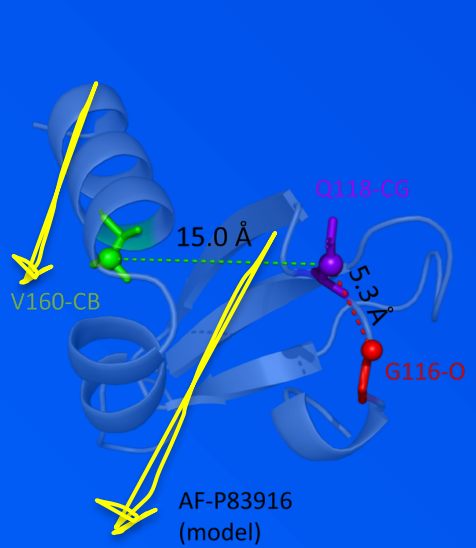
Global Distance Test, GDT_TS score

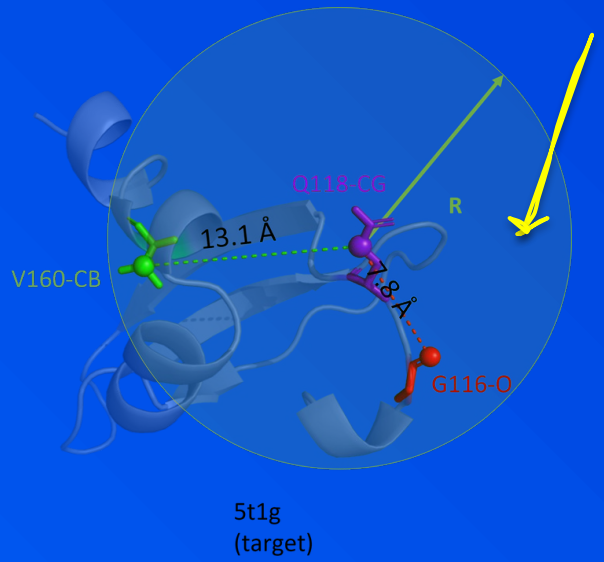
Superimposition-independent measure of similarity between two protein structures with known amino acid correspondences (e.g. identical amino acid sequences)

Fraction of conserved distances between atoms of reference and model structures

Conserved distance for threshold 2 Å

Non-conserved distance for threshold 2 Å





Threshold for considering distances

Zemla A et al. (1999) Proteins, 37(S3):22–29.

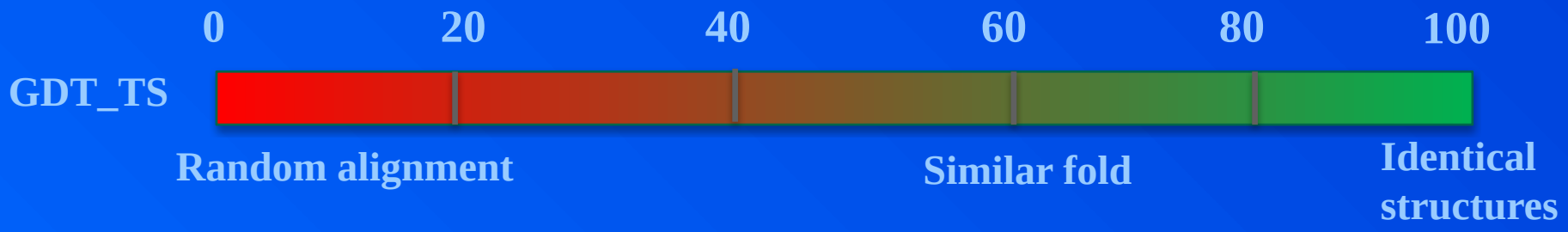
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Comparing Tertiary Structures

Global Distance Test, GDT_TS score

Superimposition-independent measure of similarity between two protein structures with known amino acid correspondences (e.g. identical amino acid sequences)

Fraction of conserved distances between atoms of reference and model structures



Zemla A et al. (1999) Proteins, 37(S3):22–29.

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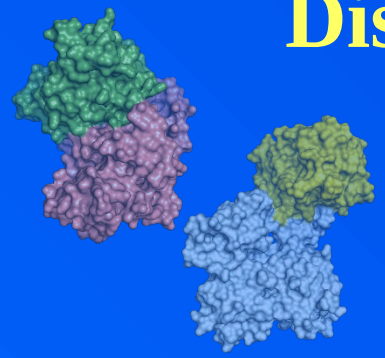
Comparing Quaternary Structures

Similar proteins

Obvious correspondence between residues



Dissimilar proteins



Vague correspondence between residues

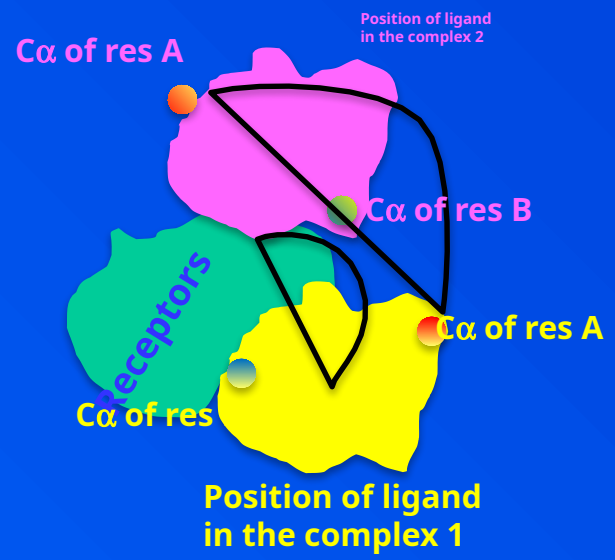
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Comparing Quaternary Structures

Similar proteins

Ligand RMSD, *L*-RMSD

align entire structure of receptors

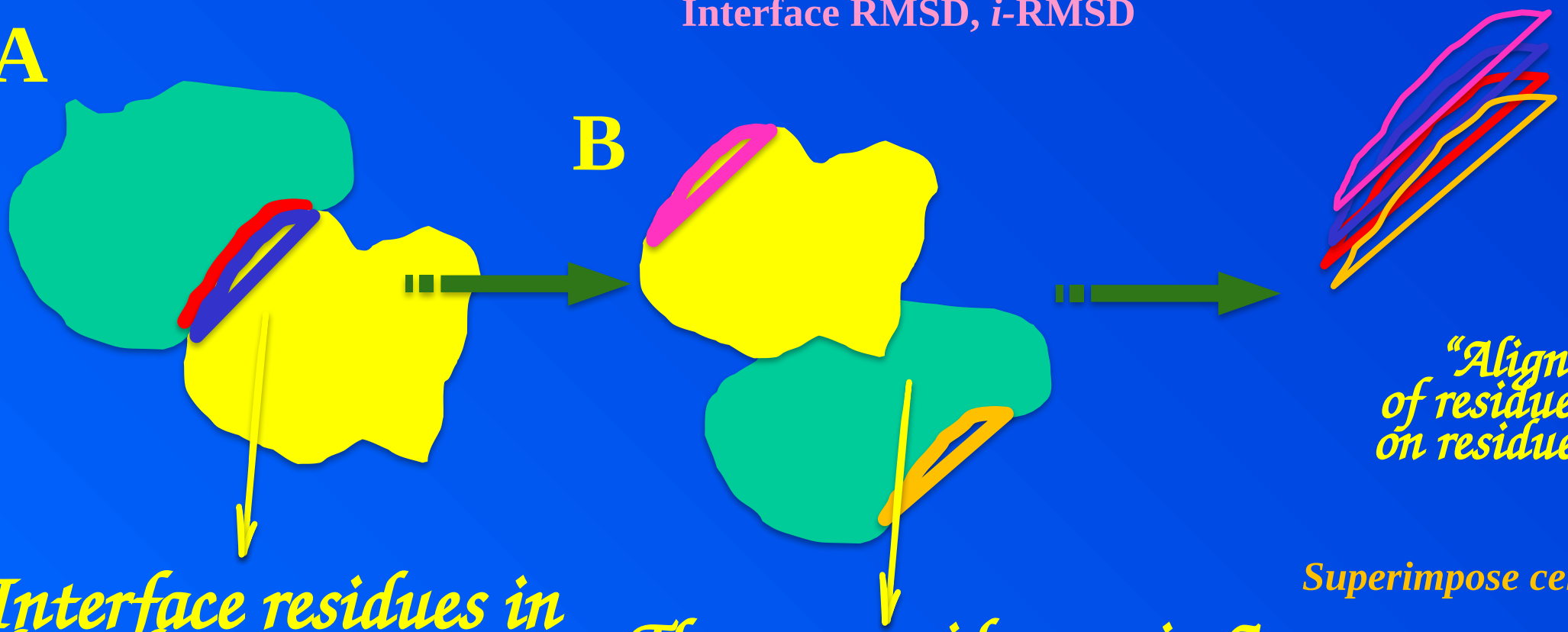


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Comparing Quaternary Structures

Similar proteins

Interface RMSD, *i*-RMSD



Interface residues in the reference structure

The same residues as in A in the docking model

"Alignment" of residues from B on residues from A:

Superimpose center of mass

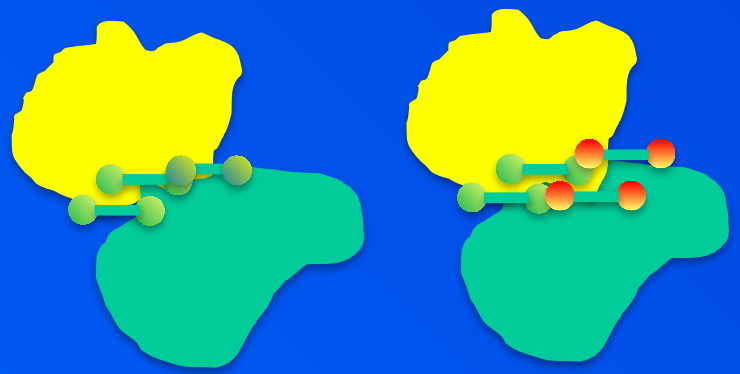
Rotate to minimize distances

Comparing Quaternary Structures

Similar proteins

Native complex

Docking model



Lensink and Wodak (2013), Proteins, 81, 2082-2095

Fraction of native contacts

$$\frac{N_{green}}{N_{green}} \quad N_{blue}$$

Fraction of non-native contacts

$$\frac{N_{red}}{N_{red}} \quad N_{green}$$

$$f_{nat} + f_{op} \neq 1$$

CAPRI criteria

Quality category	Condition
High	$f_{nat} \geq 0.5$ and (L-RMSD ≤ 1.0 Å or i-RMSD ≤ 1.0 Å)
Medium	$f_{nat} \geq 0.3$ and (1.0 < L-RMSD ≤ 5.0 Å or 1.0 < i-RMSD ≤ 2.0 Å)
Acceptable	$f_{nat} \geq 0.1$ and (5.0 < L-RMSD ≤ 10.0 Å or 2.0 < i-RMSD ≤ 4.0 Å)
Incorrect	$f_{nat} < 0.1$ and (L-RMSD > 10.0 Å and i-RMSD > 4.0 Å)

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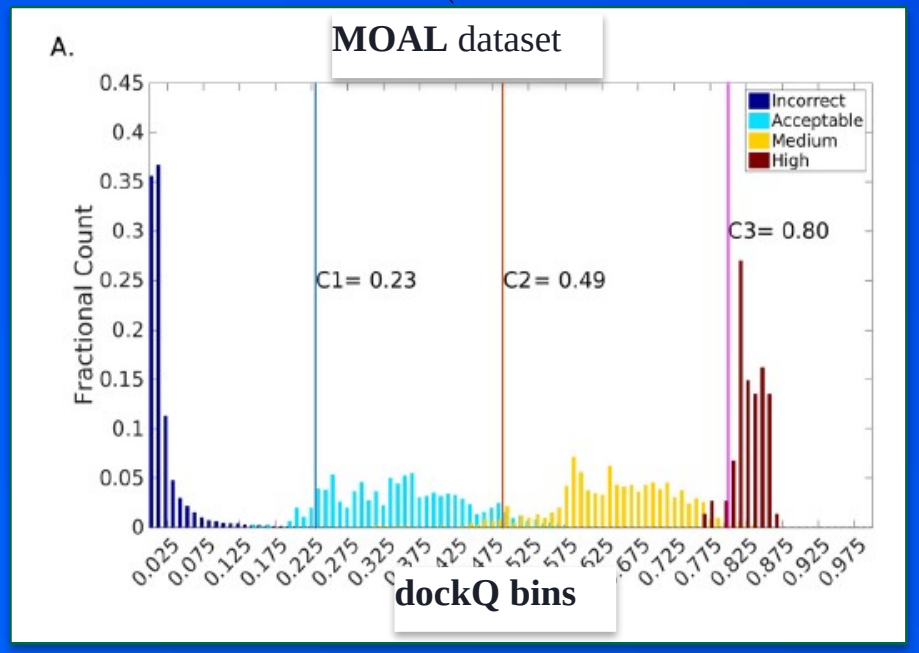
Comparing Quaternary Structures

Similar proteins

dockQ score

Combines three CAPRI criteria (f_{nat}, L-RMSD, and *i*-RMSD) into a single score

$$\text{dockQ} = \frac{1}{3} \times \left(f_{nat} + \frac{1}{\left(1 + \left(i\text{-RMSD}/1.5\right)^2\right)} + \frac{1}{\left(1 + \left(\text{L-RMSD}/8.5\right)^2\right)} \right)$$



Quality category	Condition
High	**dockQ > 0.80**
Medium	**0.49 < dockQ < 0.80**
Acceptable	**0.23 < dockQ < 0.49**
Incorrect	**dockQ < 0.23**

Basu and Wallner (2016), PLoS ONE, 11, e0161879

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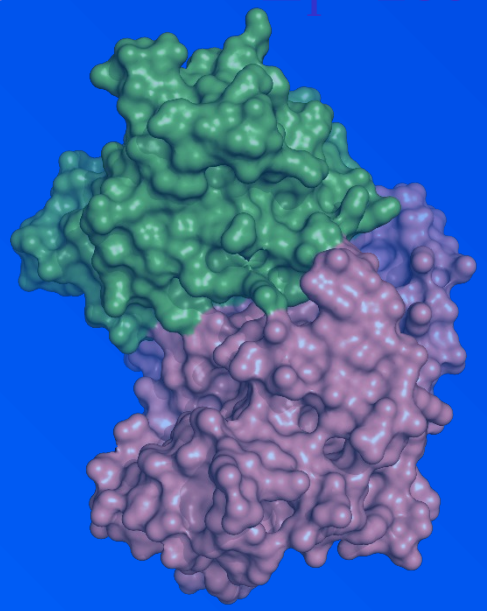
Comparing Quaternary Structures

Dissimilar proteins

Interaction RMSD, *i*RMSD

1ATN

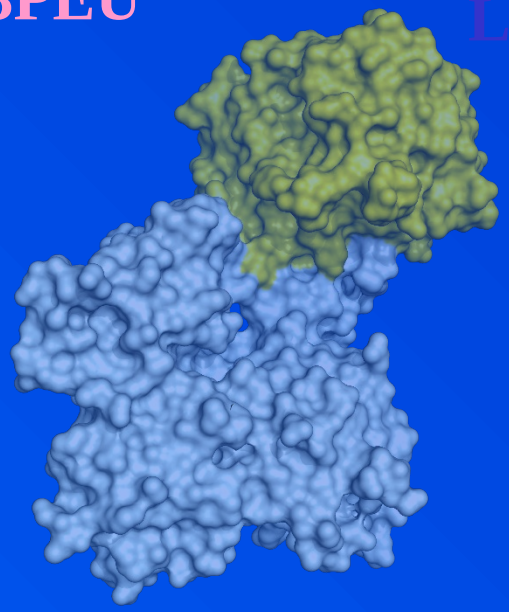
L_1= 260 aa



R_1= 373 aa

3PEU

L_2= 188 aa



R_2=297 aa

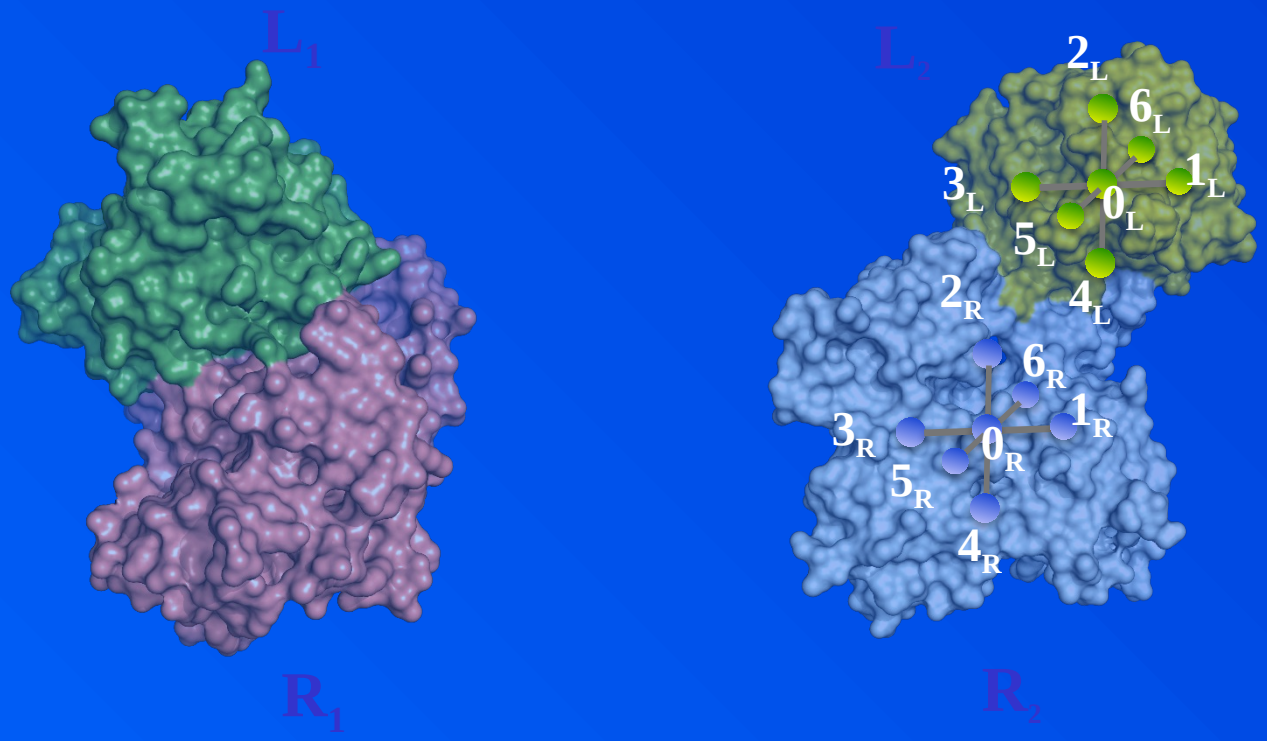
Aloy et al ., JMB (2003), 332: 989-998

Stockholms universitet

Comparing Quaternary Structures

Dissimilar proteins

Interaction RMSD, *i*RMSD



Aloy et al ., JMB (2003), 332: 989-998

Stockholms universitet

Comparing Quaternary Structures

Dissimilar proteins

Interaction RMSD, *i*RMSD

Step 1



Aloy et al ., JMB (2003), 332: 989-998

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Comparing Quaternary Structures

Dissimilar proteins

Interaction RMSD, *i*RMSD

Step 2



Aloy et al ., JMB (2003), 332: 989-998

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Comparing Quaternary Structures

Dissimilar proteins

Interaction RMSD, *i*RMSD

Step 3



***i*RMSD:**

RMSD between two sets of recorded positions

0_{RL} — 0_{RR}, 0_{LL} — 0_{LR}

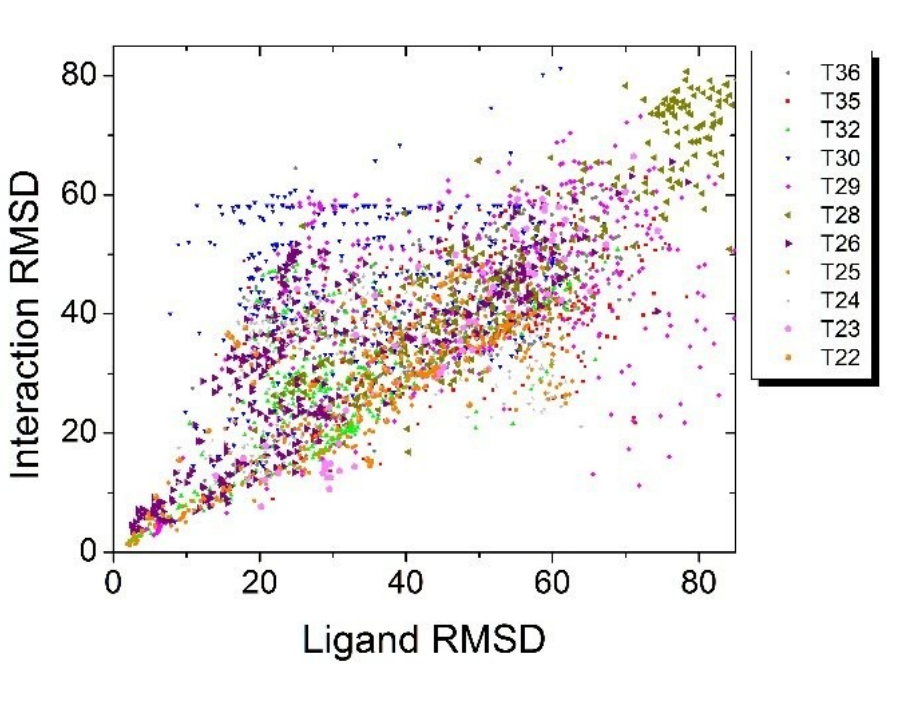
6_{RL} — 6_{RR}, 6_{LL} — 6_{LR}

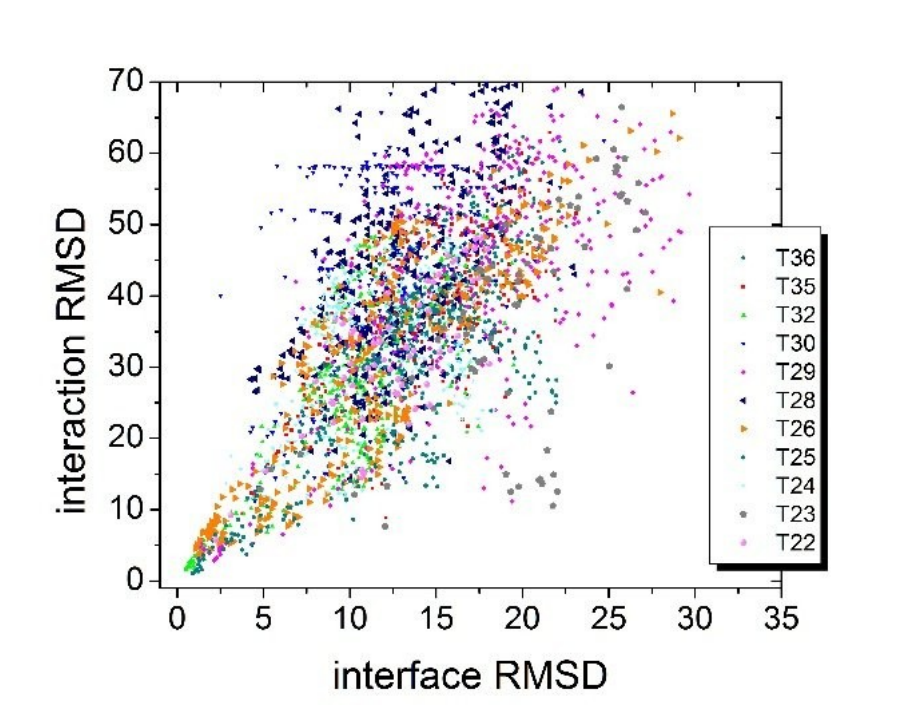
Aloy et al ., JMB (2003), 332: 989-998

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Comparing Quaternary Structures

*i*RMSD vs L-RMSD/*i*-RMSD





Kundrotas et.al., (unpublished)

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Modeling Tertiary Structure

De novo (ab initio)

Comparative (homology and threading)

Evolutionary (contacts)

Artificial Intelligence

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Modeling Tertiary Structure

De novo (ab initio, free)

Main assumptions

Non-covalent interactions between atoms of a protein are essential for determining protein tertiary structure

Protein fold into a structure with the lowest free energy (energy funnel)

Advantages

Most rigorous description of protein folding

Pitfalls

Can be applied to small proteins only (<100 aa)

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Modeling Tertiary Structure

Comparative (homology, template-based, threading)

Main assumption

Two sequences are similar → Their structures are similar too

Advantages

Computationally inexpensive → Possibility of large-scale calculations

Pitfalls

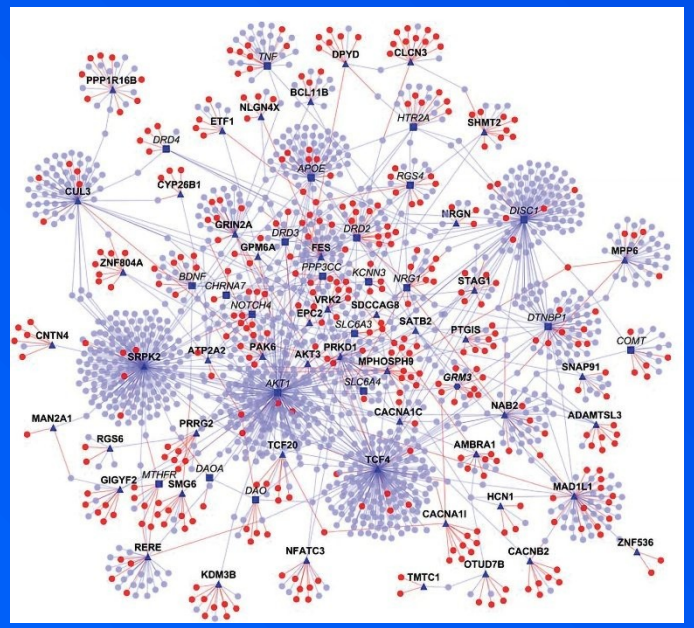
Existence of templates (previously determined structures → Completely new structures (folds) cannot be modeled

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Modeling Quaternary Structure

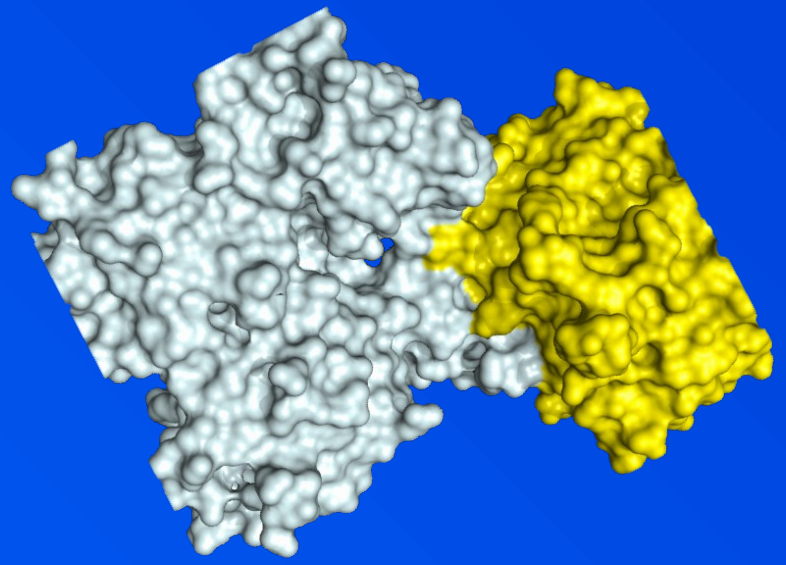
Predicting protein-protein interactions is a dual problem

Do sequences A and B interact ?



IF?

What is 3D structure of AB complex ?



HOW?

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Modeling Quaternary Structure

Template free (ab initio) docking

Template-based docking

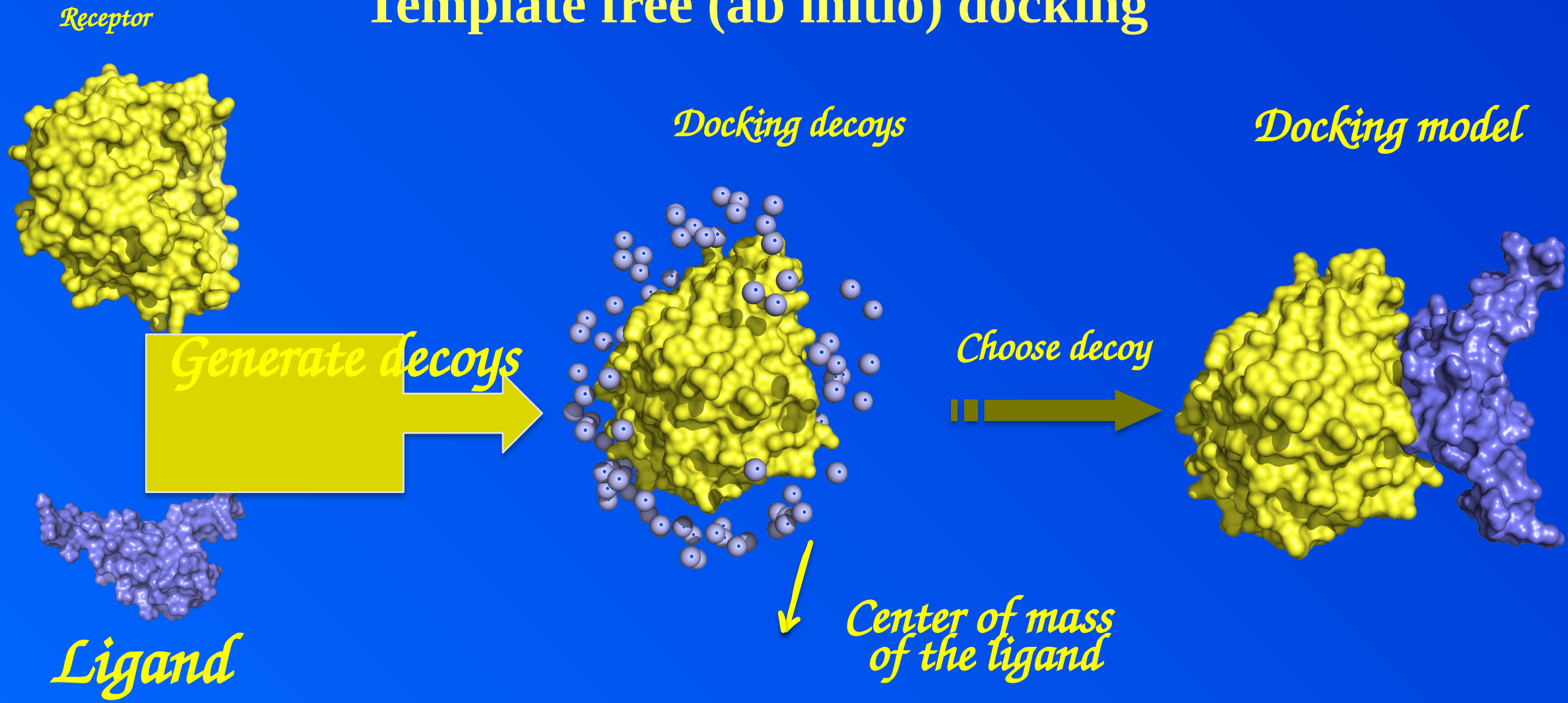
- **Sequence-based similarity**
- **Structure-based similarity**

Deep-learning (Artificial Intelligence)-based docking

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Modeling Quaternary Structure

Template free (ab initio) docking



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Modeling Quaternary Structure

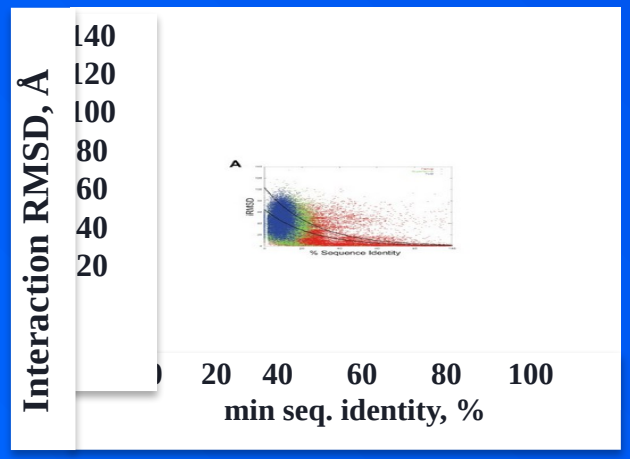
Docking by sequence similarity

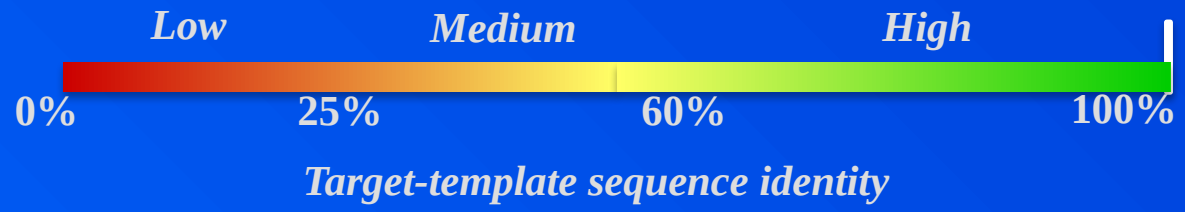
Single proteins

Two sequences are similar (homologous) → Their structures are similar too

Protein-protein complexes

Two interacting sequences A and B are similar to two other sequences A′ and B′ forming a complex → Structure of AB complex is similar to structure of A′ B′ complex.



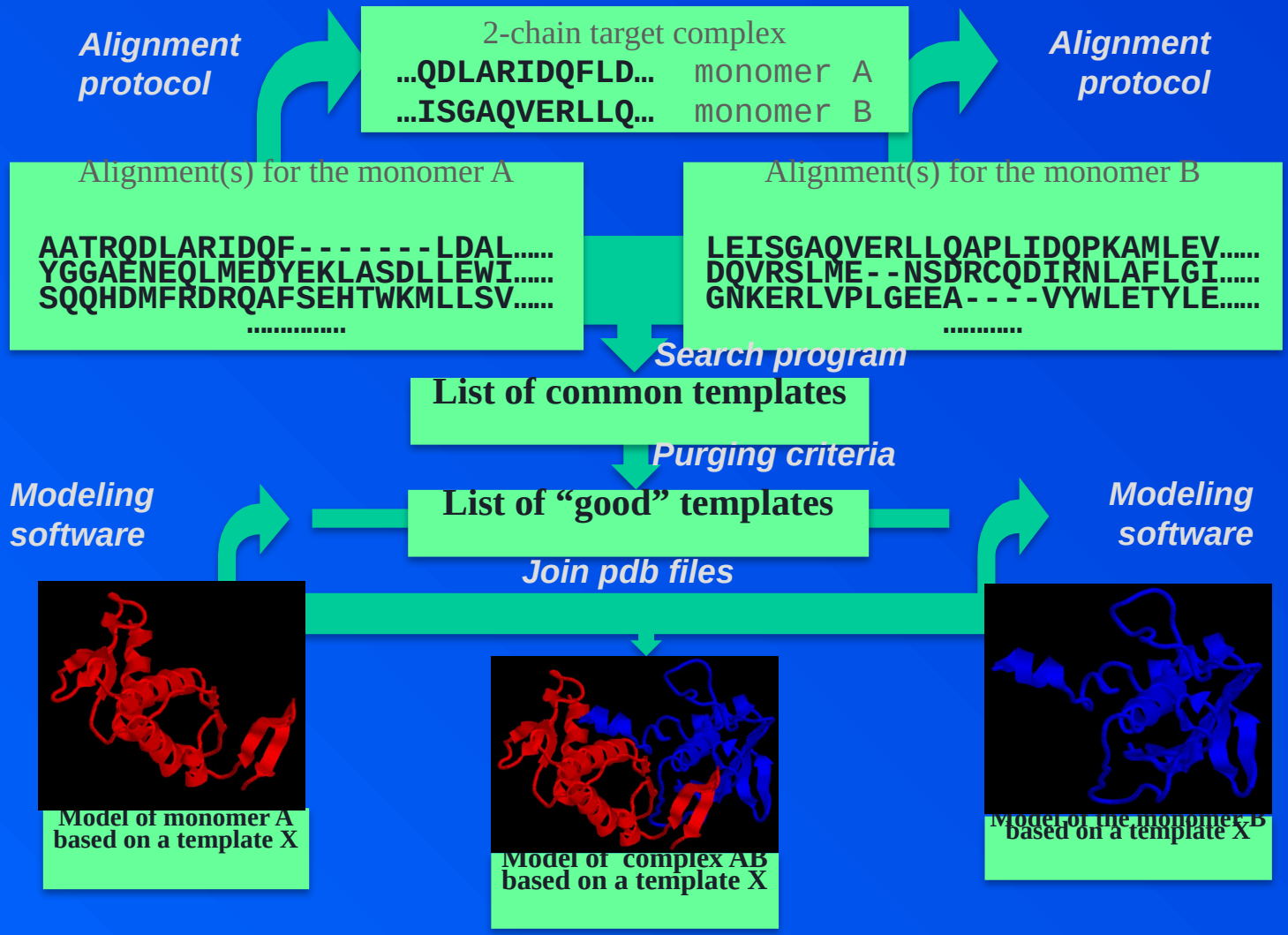


Aloy et al ., JMB (2003), 332, 989; *Curr Opin Struct Biol, (2002) 15, 15-22*

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Modeling Quaternary Structure

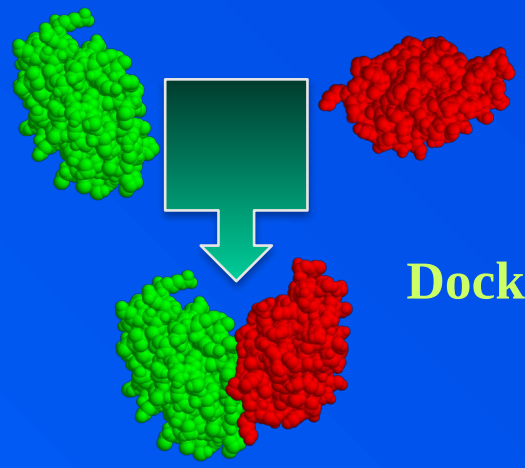
Docking by sequence similarity



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Modeling Quaternary Structure

Template free docking

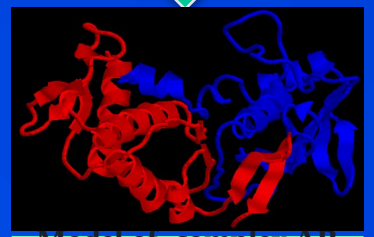


Docking by sequence similarity

2 sequences

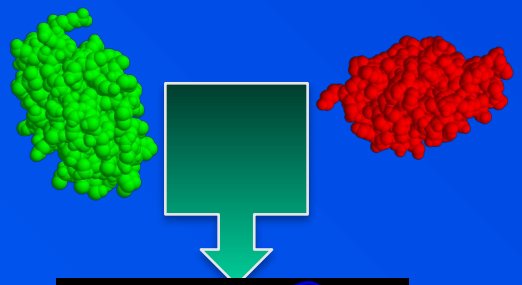
…QDLARIDQFLD… monomer A

…ISGAQVERLLQ… monomer B



Model of complex AB based on a template X

Docking by structure similarity



Model of complex AB based on a template X

Sinha, Kundrotas & Vakser (2010) Proteins, 78:3235

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Classifying Tertiary Structures

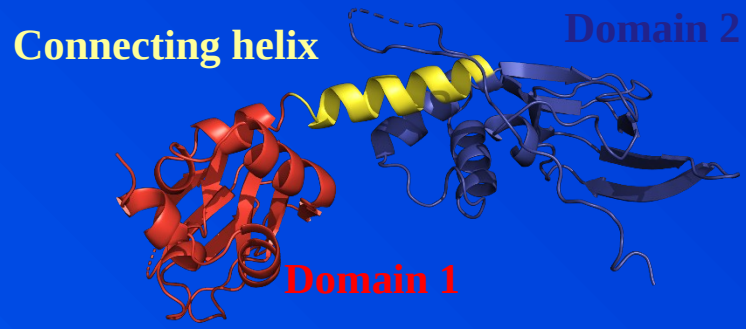
Domains

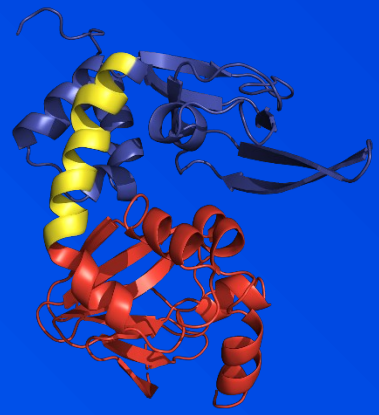
Forms a compact three-dimensional structure

May fold independently from the rest of structure

Performs distinct function

Domain arrangements may differ





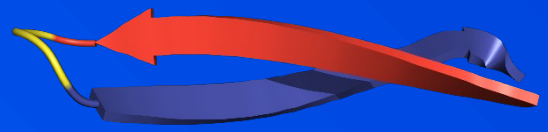
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Classifying Tertiary Structures

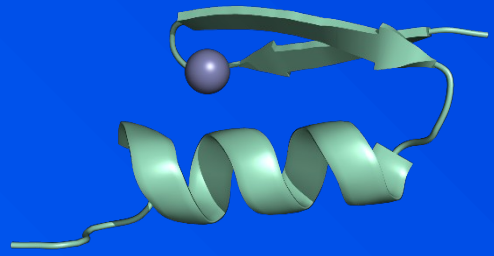
Motifs

Simple combinations of secondary structure elements that frequently occur in protein structure

β-hairpin



Zinc finger

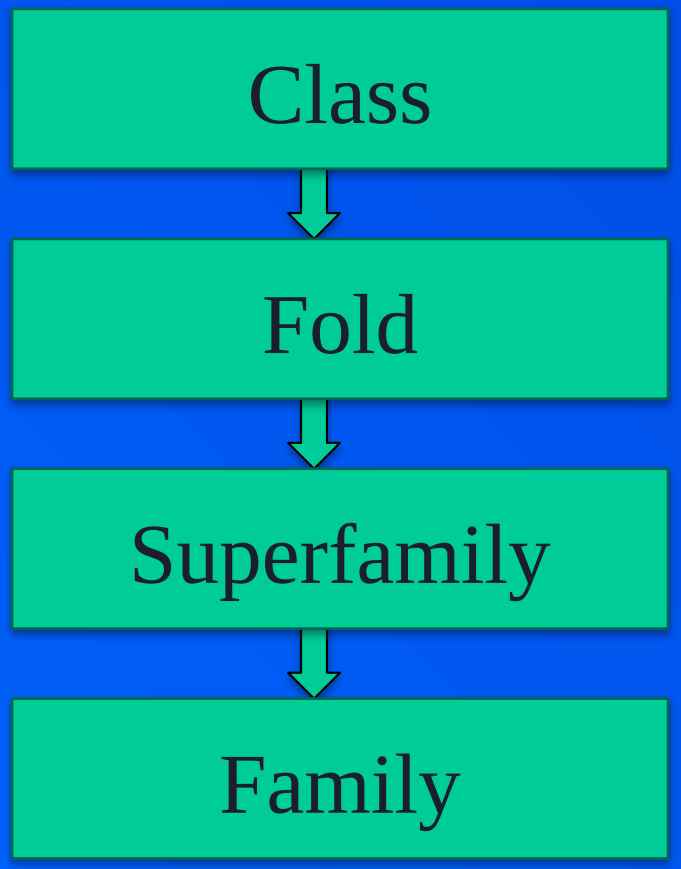


Many others

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Classifying Tertiary Structures

Generic classification scheme



Overall composition of secondary structure elements within a domain

Number, arrangement, and connectivity (or topology) of secondary structure elements

Domains with similar folds and usually similar functions with no detectable sequence similarity

Domains within a superfamily with significant sequence similarity

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Classifying Tertiary Structures

Older databases

SCOP

http://scop.mrc-lmb.cam.ac.uk/legacy/

Last update June 2009

38221 PDB entries; 110,800 domains

CATH

https://www.cathdb.info/

Last update July 2019

150,238 PDB entries; 500,238 domains

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Classifying Tertiary Structures

Newer databases

ECOD

http://prodata.swmed.edu/ecod/

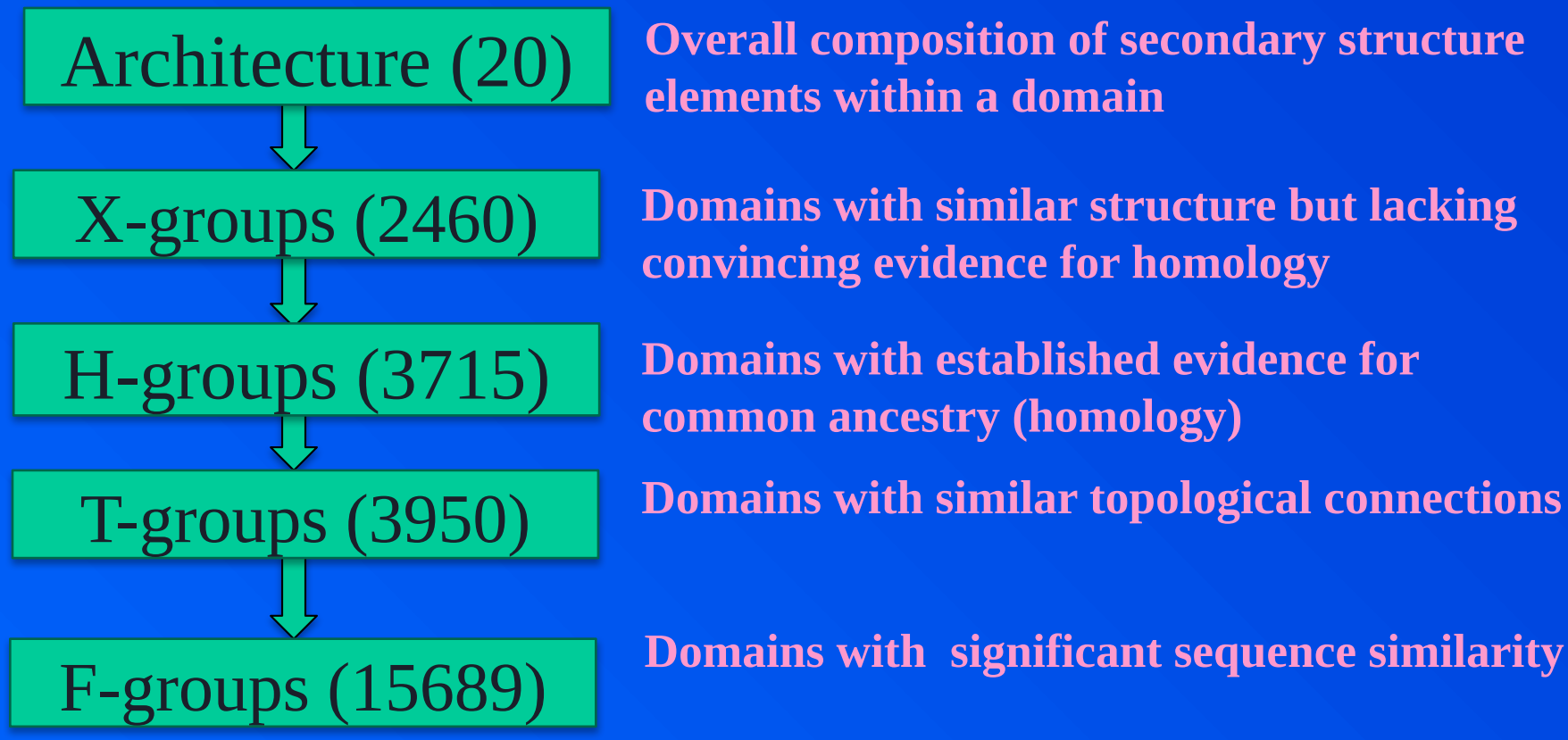
Last update October 2020

168,734 PDB entries; 789,634 domains

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Classifying Tertiary Structures

ECOD classification hierarchy



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Topics covered

Different levels of protein structures

How to compare protein structures

How to predict protein structures

How to classify protein structures